Flock: Forecasting How Busy a Venue Will Be, and Counting How Many People Are There

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October 2026. A snapshot before the early-November retrain; to be updated with its held-out results.

Abstract

The venue card in Flock, an app for planning nights out, answers one question: how busy will this place be when the group arrives? I asked whether a venue's own recent live readings beat the popular-times chart for one venue at one hour. A serving path corrects each venue's weekly curve with a trailing offset and an age-weighted nowcast of its latest reading, with every weight fitted on earlier days. Scored against a commercial provider's live busyness estimates, relative 0 to 100 readings rather than headcounts, on 119,939 readings its weights never saw (9 September to 8 October 2026), it named the exact crowd level (of five) 54.9% of the time, against 30.2% for popular times, with a mean absolute error 31.8% lower. The gain comes from the venue's latest live reading: one hour ahead, with a reading from the hour before, the served number lands on the exact level 77.0% of the time, and a simplified rule built on this finding now serves production. Two to 24 hours ahead, when no fresh reading exists, the corrected curve still names the exact level about 3 points more often than popular times. Against its own live readings the provider's weekly curve is a poorly calibrated forecast (slope 0.50, r 0.36). Repeating the analyses on 5.3 million hourly counts from 315 public counting sites in seven countries, under a plan fixed before any data were downloaded, shows that the provider-specific findings do not carry over: counted crowds follow their curves far more closely (slope 0.92, r 0.79), and the raw hour-old count loses to the curve in 9 of 15 datasets, though where the counts are people inside a place it wins by a wide margin, as on the provider's readings. The latest reading helps everywhere as a departure carried onto the curve, which gives the lowest one-hour error in 14 of 15 datasets, 33% below the curve when pooled (an exploratory comparison). To supply real counts, I built Owl, a 718,550-parameter network that counts people on a 160 × 120 thermal camera in 14 to 16 ms on one Raspberry Pi 5 core. Trained on physics-based generated frames and licensed real data, it counts 82 to 96% of frames exactly on five of six real benchmark sets used in its development, where the heat-threshold rule it replaced scores 9 to 77%, and 94.3% on a sealed set of 300 office frames of one or two people, against 40.7% for the rule; on cameras unlike those it was trained on, it transfers unevenly. A field study in a working venue comes next. I will update this snapshot with the held-out results of a pre-registered retrain in early November 2026, the project's first test whose decision rule and held-out window were fixed before its data existed.

Keywords: crowd prediction; popular times; nowcasting; gradient boosting; thermal imaging; people counting; on-device inference; privacy by design

Outline. Sections 1 to 6 cover the forecast: the question, the data and what goes into the model, how the served number is made, the results, the pre-registered retrain and the conclusion. Sections 7 to 13 cover Owl, the people counter. Section 14 records how both were built, Section 15 gathers the limitations of both studies, and Section 16 sets out what comes next. The appendix lists the choices made after seeing data.

PART I

Forecasting how busy a venue will be

A crowd number that watches each venue's own recent readings, tested on a month of live readings its weights never saw.

1 Introduction

A group deciding where to go tonight wants one fact about each option: whether it will be busy when they arrive. Consumer maps mostly answer with the popular-times chart, the venue's typical level for that weekday and hour, averaged over a trailing window and drawn as a bar chart, sometimes with a coarse live indicator beside it. Möhring et al. studied the Google form of it as a behavioural signal [1]. It is a strong prior. By construction it cannot see the specific evening: the rain, the game two blocks away, the holiday, the week the place became popular.

I build Flock, an app for planning nights out with friends. Its venue card shows one integer from 0 to 100 and one of five words: *Quiet*, *Not Busy*, *Steady*, *Busy* and *Packed*. Whether that number is right decides whether a plan works. My research question is this: on the venue-hours a production system actually serves, can a system that watches venues over time beat each venue's own typical pattern, and by how much?

I tested three hypotheses.

- **H1.** Correcting the popular-times curve with the venue's own recent live readings reduces error against held-out live readings, on both the point scale and the five-level scale.
- **H2.** A model that predicts how an hour departs from the curve, using context such as weather, events and calendars but no recent history, beats the curve on cities it has never seen.
- **H3.** The value of a live reading decays within hours, so the weight it deserves falls with its age.

Prior work sets the bar. BysGNN [2] forecasts hourly venue visits six hours ahead from the previous 24 hours of visits at every venue in a city, a live feed a consumer app rarely holds. Against a Historical Average baseline, which its authors liken to the popular-times graph, its MAE is 40% to 54% lower across five cities at its 400-venue scale; against last week's value at the same hour the gain is 14% to 19% (both computed from its tables). Timokhin et al. [3] estimate hourly venue popularity from venue attributes and compare popular-times data with WiFi counts. Beating a trivial baseline is not guaranteed: in a beach-occupancy study a feature-based random forest and SARIMA both had higher MAE than naive persistence [4], and in weekly foot-traffic forecasting a simple weighted average beat an LSTM [5]. So every candidate in this project is scored against the number production serves and against persistence of the latest reading, the standard naive benchmark [6], on the same rows.

The main finding is that a venue's freshest live reading is worth more than any correction to its typical-week curve. Over the 30-day test, adding the latest reading lifts within-10 accuracy from 31.2% to 55.2%, and where the venue was read an hour earlier the served number lands on the exact crowd level 77.0% of the time (52.9% where the value had changed since that reading, against 29.1% for popular times). A venue's departure from its curve correlates 0.74 from one hour to the next and 0.26 across two hours, so a reading's value is concentrated in its first hour. Production now uses readings up to an hour old at full weight and the corrected curve otherwise. These are one-hour-ahead results; from two to 24 hours ahead the corrected curve keeps a smaller, steady lead over popular times (Section 4.5). A replication on 5.3 million public counts shows where the provider's readings differ from counted crowds, and that the latest reading still helps there as a departure from the curve (Section 4.10).

The labels throughout are a commercial provider's relative busyness readings, scaled 0 to 100 within each venue. They do not count people.

My contributions are:

- A serving path, in production since 26 September 2026, that adds a trailing per-venue offset and an age-weighted nowcast to the venue's curve, with every weight fitted strictly before the rows that score it. On 119,939 live readings its weights never saw, it names the exact crowd level 54.9% of the time, against 30.8% for the configuration it replaced and 30.2% for popular times, and cuts the popular-times baseline's MAE by 31.8%, all measured against the provider's live estimates. Scoring it against persistence baselines locates the gain in the venue's latest reading, and a simplified rule built on that finding has served production since 13 October.
- Three properties of typical-week curves, measured on the provider's readings: as a forecast the curve is poorly calibrated, with observed busyness rising about half a point per point of curve; a venue's departure from its curve stays correlated for one to two hours; and within-10 accuracy partly rewards predictions that land on the labels' 5-point lattice. On real counts the curve is close to calibrated and departures last longer (Section 4.10).
- An external replication on 5,321,507 hourly real counts at 315 public sites in seven countries, with its analysis plan fixed before any data were downloaded, separating what holds for counted crowds from what describes the provider's estimates.
- A gradient-boosted delta model, evaluated on three cities it never trained on, that takes R^2 above zero at single-venue, single-hour granularity, with a complete account of what goes into it.
- An evaluation apparatus that outlasts any one model: a ship gate against production as served, trainer-server parity tests, a dataset that records its own provenance, and a retrain test whose rules were committed before its data existed.

2 Data and what goes into the model

2.1 The corpus

At the training run the corpus held 3,912,357 rows: 3,454,955 weekly and 457,402 realtime, across 34,784 venues in 34 cities, collected between 2026-03-10 and 2026-05-18. The weekly rows are the busyness provider's typical week for each venue: 168 hourly values on a 21-point lattice of multiples of 5. These

rows define the curve. Realtime rows were scarce: 457,402 observations over 18,442 venues is 25 per venue across 168 weekly slots.

Context comes from four further sources: weather at the hour from OpenWeather; ticketed events from Ticketmaster listings; category, rating, review count, price level and place types from a places listing; and holiday, school-break and special-night calendars across 29 country and city scopes. A venue owner's own live report enters training at weight 0.10. The venue's location is used to look up weather, daylight and neighbours and is never a feature.

Before unique keys enforced one weekly row per (venue, weekday, hour) and one realtime row per (venue, date, hour), I found and repaired three faults. The hour column was an array index in 3,536,360 rows, so slot 18 held the venue's midnight. There were 491,100 duplicate rows. And 63% of rows carried an impossible month-season pair. Every accuracy figure in this paper comes from after the repair.

2.2 Live readings and their provenance

The 457,402 spring realtime rows were collected before provenance was recorded. They cannot be separated into live readings and provider forecasts, and they stay marked *unknown*. Since 2026-09-01, and hourly from 3 September, a collector reads every tracked venue's live value and stores one row per venue per hour, stamped with the venue's local clock and a provenance label (live or provider forecast). "Live" in this paper means these provenance-verified readings.

About 1,380 Pennsylvania venues are scheduled every hour, at seven minutes past, and Miami venues at two fixed hours a day. Since 2026-09-25 the full sweep finishes inside the hour. Verified readings per venue were thin at first: on 2026-09-06, 9,339 rows across 1,228 venues carried verified live provenance, a median of six each.

2.3 The delta model and its inputs

The delta model predicts how an hour departs from the venue's curve. It reads 106 features in eleven families, listed in Table 1. The venue's own curve is the anchor and is refused as an input, and every column that would leak the label is excluded, following Kapoor and Narayanan's taxonomy [7]. The learner is XGBoost [8] with 800 trees at depth 8 and learning rate 0.01. The shipped model, v2.6.0-starling, was trained on 1,934,988 rows from 30 cities, with weekly rows at weight 0.05 and spring realtime rows (provenance unknown) at 1.0. Coordinates, venue identifiers, the venue's own curve and every provenance column were excluded. Miami, Tokyo and Barcelona never enter its training; they are the held-out cities, following the practice of holding out whole spatial blocks [9]. The corpus's 34th city, Beijing, holds no rows in its training frame.

The model's score is rebuilt by a fixed chain:

model score = round(clip(curve + 0.5 × offset + clamp(predicted departure, ±50), 0, 100))

In model mode a 40-knot monotone display map, fitted to v2.6.0's output, then follows; rounded scores outside 25 to 65 move one point toward the nearer end of the scale; and a nowcast with weights fitted on the model's own output blends in the latest reading.

Table 1. Every input of the delta model v2.6.0-starling (106 features), by family, with its source at serving time. The last three rows are not inputs of v2.6.0: the venue's curve is the anchor, and the families marked † are built for the pre-registered retrain.

Family	Features	Source at serving
Time (19)	hour, weekday, month; sine and cosine of each; weekend; Friday or Saturday after 18:00; lunch, dinner, late night, morning; four season flags	The venue-local clock of the hour being scored
Daylight (3)	daylight hours, hours after sunset, after-sunset flag	Solar estimate from the venue's latitude and the month
Holidays (6)	holiday, school break, holiday eve, special night, special-night boost and suppress weights	US federal holidays, US school-break windows, and 31 national holiday calendars with 1,351 dated special nights over 29 scopes (2025 to 2028)
Weather (18)	temperature, humidity, wind, rain flag; nine condition groups; temperature anomaly against a latitude-band by month climate table; four weather interactions	Current conditions keyed to about 1 km, cached 30 min; a missing reading becomes the climate norm
Events (16)	any event within 2 km; nearest event's attendance and its log, and distance; its type as five flags (arts, family, music, sports, other); count, total attendance and its log within 2 km; large event; event × weekend, dinner, bar	Ticketed events under way at the scored hour within 2 km; a failed lookup enters as no event
Category curves (2)	typical value of the venue's category at this weekday and hour; the same, refined by price and rating tier	Tables fitted on the training corpus, shipped in the model's metadata
Neighbours (2)	log count of venues with a curve within about 830 m; their mean curve value at this weekday and hour	Stored weekly curves of nearby venues, the venue itself excluded
Venue (5)	category (13 classes), price level, rating, review count and its log	The venue's places listing; missing values take the corpus median, or 0 reviews
Place types (30)	one flag for each of the 30 most common place types	Same listing, from the venue's first three types
Feedback (4)	mean crowd report, log report count, has-report flag, mean error of past predictions	Per-venue aggregate of verified in-app reports from the last 28 days
Row type (1)	realtime or typical-week row	Always 1 at serving
Own curve (0)	refused as an input	The busyness provider's typical week, smoothed 0.6 / 0.2 / 0.2
Live readings† (5)	last earlier reading's departure from the curve and its age; trailing offset and its count; the curve one hour earlier	The venue's own live readings strictly before the scored hour
Sports games† (6)	a tracked team plays today; number of games; evening game; home game; distance to the home arena; arena within 10 km	Public sports schedules, for venues within 60 km of a tracked arena

I counted how the shipped trees use these families from the ONNX file: 167,758 decision splits in the 800 trees. Venue attributes take 20.0% of splits and weather 19.5%. Log review count is the most split-on feature (9.3%), and the two neighbour features are the second and third (7.1% and 6.7%). A split count favours continuous features with many cut points over binary flags, so it ranks families only roughly. Two facts are exact. Ten features are never split on and cannot affect the output: the four season flags,

the holiday flag, the four feedback aggregates and the family-event flag. And month carries 274 of the 800 first splits, a reflection of a corpus collected in March, April and May, which the retrain answers by dropping the month family.

No input identifies or locates a person. Latitude reaches the model only through daylight, the climate band, the special-night rule and the neighbour box. In-app feedback enters as four per-venue aggregates, which the shipped model never splits on.

The ship gate. A candidate model must pass two gates before it can serve. The point gate asks it to beat the curve on the held-out cities by 5 MAE or 0.10 R^2, not regress MAE, and match the incumbent's own within-10 on the same rows. The band gate asks it to beat production *as served* on at least 1,000 held-out live readings over 5 or more dates, with date-block confidence bounds; its criteria are listed with the pre-registered retrain below. The server fails closed: an artifact that did not pass cannot be loaded.

Reproducibility. Following common reproducibility practice [10], training pins the device, thread count and seed, and a test confirms that two runs on the same data produce bit-identical predictions; the model is served in-process through ONNX Runtime [11]; and each feature family is computed by the trainer and the server from one definition, with tests that report zero disagreements row by row.

3 How the served number is made

3.1 What the curve gets wrong

Before designing anything, I measured the popular-times baseline against verified live readings. Three findings set the design.

The curve is a poorly calibrated forecast. I joined every verified live reading to its own venue's curve value for the same weekday and hour, giving 9,903 paired rows. The gap inverts with the level of the curve (Figure 1). In hours whose curve value is at most 25% of the venue's daily peak, observed busyness sits 15.60 points above the curve (1,162 rows). In hours above 75% of the peak it sits 13.56 points below (5,746 rows). Regressing observed busyness on the curve gives observed = 18.82 + 0.466 × curve, with r = 0.372: each point of curve is worth about half a point of observed busyness, and the curve explains about 14% of the variance. The readings themselves are not narrower than the curve; they are more spread out (standard deviation 33.2 against 26.4), and the slope is low because the correlation is low. The best forecast from such a curve shrinks it toward the mean, the regression effect in prediction [12, 13]. The 30-day test window repeats the pattern: observed = 18.62 + 0.497 × curve, r = 0.360, with quiet hours 11.5 points above the curve and busy hours 13.9 below. A correction must therefore be able to move quiet hours up and busy hours down, which one global offset cannot do.

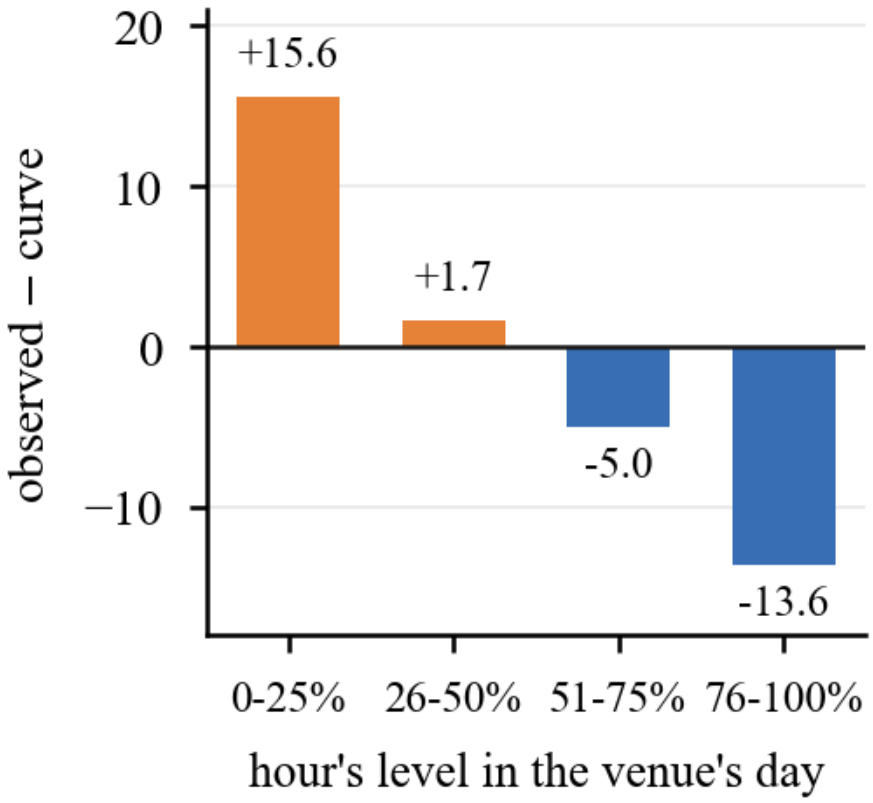


Figure 1. Observed minus curve, by the hour's level in the venue's own day (9,839 of the 9,903 paired rows). Quiet hours read above the curve and busy hours below it.

A venue's recent state persists, then fades. In the first week of hourly collection (1 to 8 September) the provider's live value was sticky: 80.9% of consecutive-hour readings of a venue were identical, and a venue's deviation from its curve correlated 0.815 across one hour, 0.487 across two and 0.203 across three (Figure 2). Used as the prediction for the next hour, a reading one hour old landed within one crowd level 92.0% of the time and within 10 points 85.9% (3,165 readings). Nothing in a venue's typical week carries this information, so the latest reading should enter the served number directly, at a weight that decays on the same timescale. The stickiness did not last. Identical consecutive readings fell to 69.3% from 9 to 25 September and to 44.8% and 47.5% in the two periods after, and over the 30-day test window the deviation correlates 0.739, 0.255 and 0.158 at one, two and three hours. Whether the early stickiness came from how often the provider refreshed its value or from when the collector read it cannot be told from these data.

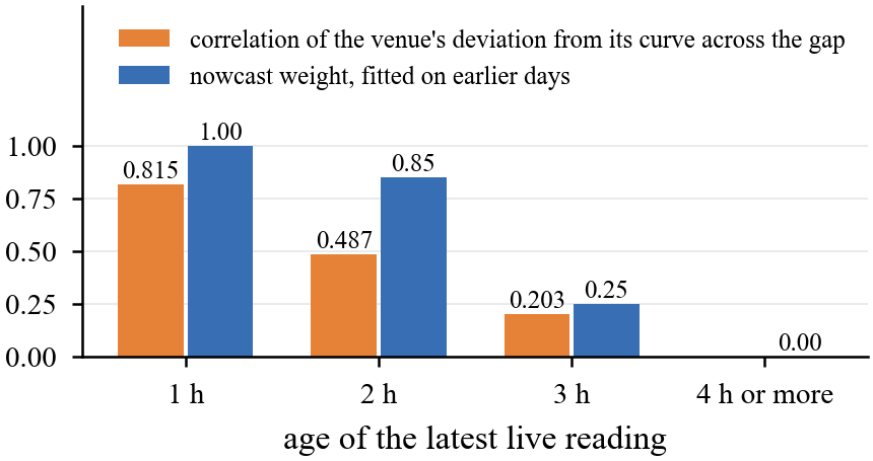


Figure 2. How a venue's deviation from its curve persisted over 1 to 8 September, against the nowcast weights fitted on 1 to 5 September. The fitted weights fall with age, more slowly than the correlation at two hours. Over the 30-day test the correlation at two hours fell to 0.26.

The within-10 metric rewards the grid. The labels lie on a lattice of multiples of 5, and so does the stored curve. In a forward split (894 venues, 3,930 later readings), MAE was best at an offset weight of 0.5 (25.46, against 26.21 with no offset), while within-10 was highest at the two weights whose predictions most often land on the grid, 1 (34.55%) and 0 (32.88%). On a lattice-valued dataset, within-10 partly measures grid coincidence. I therefore read it beside MAE and the level metrics, and where a gridded and a non-gridded predictor are compared on it, the comparison reports the on-grid rate or projects both to the lattice. Production's within-10 margins in Table 3, 24 points over the weekly curve and over curve plus offset, are far larger than the within-10 variation across the offset weights tested here.

3.2 The arithmetic

For a venue at a target hour, production computes the number in this order. The target hour's own reading is never an input.

1. **The weekly curve.** The venue's stored weekly value for that weekday and hour on its local clock, smoothed with its two neighbouring hours at weights 0.6, 0.2 and 0.2, rounded and clipped to [0, 100]. The popular-times baseline in the results is the stored weekly value itself, before smoothing, as a popular-times chart shows it.
2. **The trailing offset.** The median of (reading − curve at the reading's hour) over the venue's newest 20 live readings taken strictly before the target hour and within 28 days. It is used only with at least two readings and a table rebuilt within seven days, it is clamped to ±50, and it is otherwise 0.
3. **The nowcast.** The newest live reading from an earlier hour, if at most 12 hours old, pulls the number toward itself.
4. **The level.** The card shows the rounded number and its level: *Quiet* up to 20, *Not Busy* 21 to 39, *Steady* 40 to 69, *Busy* 70 to 84, *Packed* 85 to 100.

Written out:

$$\text{base} = \text{curve} + 0.85 \times \text{offset}$$

$$\text{served} = (1 - w(\text{age})) \times \text{base} + w(\text{age}) \times \text{latest reading}$$

with w = 1.0 at an age of one hour, 0.85 at two hours, 0.25 at three hours and 0 at four hours or more. The offset weight and all four blend weights were fitted by MAE on 2026-09-01 to 09-05 and served unchanged through the 30-day test. On 13 October the two- and three-hour weights were set to zero (Section 4.4). At an age of one hour the fitted weight is 1.0, so production then serves the last reading unchanged: the "repeat the last reading" baseline is the case the fit chose for fresh readings, and the offset takes over as readings age. The offset weight of 0.85 is the value this fit chose on those five days; the forward test above, on a different split, preferred 0.5. The weights were fitted while readings were at their stickiest, and Section 4.4 shows that over the month the two- and three-hour weights no longer helped.

The card also publishes a confidence: the within-15 rate measured, on the 30-day test window, for the arithmetic that produced the number. It is 42.7% for curve and offset alone, and 83.0%, 40.7% or 38.6% when a reading one, two or three hours old entered. Until 8 October it published the three-day figures (41.7%, 87.4%, 62.8% and 43.5%), and the month brought the three reading-age figures down, the two-hour figure most. Since 13 October a two- or three-hour reading no longer enters, so those numbers publish 42.7%.

This arithmetic, the online correction, runs no model, so weather, events, calendars and neighbours do not reach the number. A server with no loaded model is answered by a rule engine that reads the hour, weekday, rating, review count, price level, place types and weather; a venue with no curve of its own was answered the same way until 14 October and now gets its category's typical curve (Section 4.8). The delta model runs only in the code's *model mode*, which production has not served since 26 September. That day also took a 40-knot monotone display map out of the served path. It had sat between rounding and the level cut since 2026-08-28.

4 Results

The served number is scored on every model-served live reading from 2026-09-09, the day after its first test, to 2026-10-08, the day of the export: 119,939 readings over 30 dates at 1,754 venues (Philadelphia 63,225, Lehigh Valley 49,503, Miami 7,211). Each reading is rebuilt through the production serving code. Two persistence baselines are scored beside it: (a) the newest earlier reading if at most 12 hours old, else popular times; and (b) the reading from the hour before if there is one, else curve plus offset. The offset and nowcast weights were fitted on 2026-09-01 to 09-05, so no reading in the window informed them.

Intervals are date-block bootstraps that resample whole observation dates (2,000 resamples) [9], so each interval in Tables 2 and 3 resamples 30 date blocks. Readings of one venue on one night are correlated, and the date blocks carry that: on the exact-level margin over popular times they widen the interval about 4 times over treating readings as independent, about 16 times the variance. A two-way bootstrap that resamples venues and dates together widens the intervals by a further 10% to 17%; where it changes a conclusion, Table 3 says so. In the results tables, level agreement appears with exact level and band MAE, the mean number of levels between served and observed, beside it.

4.1 The 30-day test

Table 2. The served number and its comparators on 119,939 live readings over 30 dates (2026-09-09 to 10-08), none of which informed the offset and nowcast weights (fitted on 09-01 to 09-05). Rows (a) and (b) are persistence baselines; (b) was found on this window, so its row is not out of sample. Production and the baselines carry 95% date-block intervals. Bold marks the best value. Bias is the mean of predicted minus observed. The last row is production on its first, three-day test, for reference.

Configuration	Within one level ↑	Exact level ↑	Within 10 ↑	Band MAE ↓	MAE ↓	Bias
Production: curve + offset + nowcast	78.4% (77.9–78.9)	**54.9%** (53.9–55.7)	**55.2%** (53.9–56.2)	0.772	18.00 (17.73–18.28)	−1.24
(b) Hour-old reading, else curve + offset	**79.0%** (78.3–79.6)	54.7% (53.6–55.5)	54.9% (53.4–56.0)	**0.756**	**17.68** (17.38–18.03)	−1.16
(a) Reading up to 12 h old, else popular times	74.3% (73.4–75.0)	53.2% (52.0–54.1)	55.1% (53.6–56.2)	0.866	20.05 (19.67–20.50)	+0.48
Curve + offset, no nowcast	70.5%	34.4%	31.2%	1.058	24.55	−1.45
Weekly curve alone (popular times)	64.8%	30.2%	30.9%	1.155	26.40	+4.89
Rule engine	60.6%	22.1%	19.6%	1.287	29.63	+3.95
Before: v2.6.0 as served (map on)	58.3%	30.8%	26.9%	1.337	30.94	+13.94
Constant "Not Busy" (reference)	*75.2%*	*14.8%*	*24.5%*	*1.270*	*28.37*	*−11.82*
Production on the first test (4,183 readings, 09-06 to 09-08)	*78.8%*	*56.0%*	*53.8%*	*0.735*	*17.06*	*−0.98*

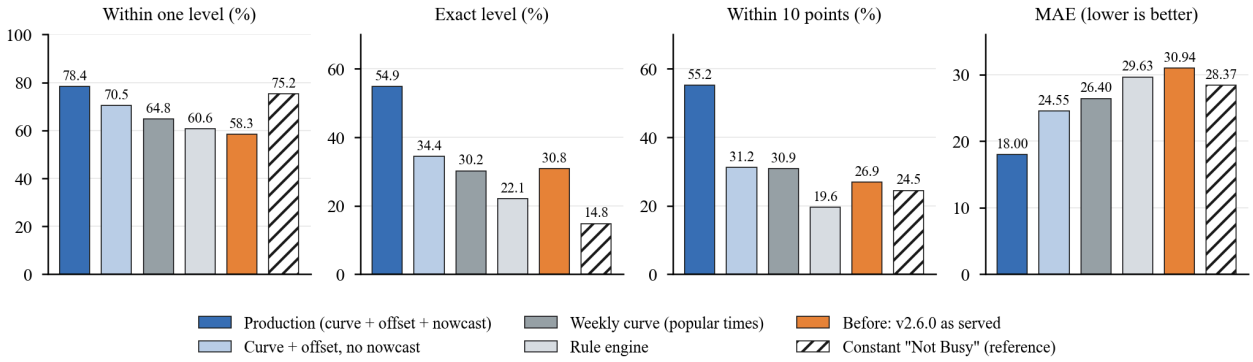


Figure 3. The served number against its comparators on 119,939 live readings over 30 dates (9 September to 8 October 2026).

Table 2 and Figure 3 give the result. Against the configuration it replaced, the delta model v2.6.0 with its display map, the served number gained 20.0 points within one level, 24.1 on the exact level and 28.3 within 10 points; MAE fell 42% and the bias fell from +13.9 to −1.2. Against the popular-times baseline, MAE is 31.8% lower and the exact level is 24.7 points higher.

The constant row sets the floor. Real venue-hours are bimodal (36.6% *Quiet* and 17.0% *Packed* over the window), and "Not Busy" sits within one level of three of the five levels, so a constant scores 75.2% within one level. It names the exact level 14.8% of the time, with a band MAE of 1.270 against

production's 0.772. On the three-day test the constant tied production on within one level; over the month production clears it by 3.1 points (1.1 to 5.1).

Table 3. Production's margin over each comparator on the same 119,939 readings, in points, with 95% date-block bootstrap intervals over the 30 dates. A negative MAE margin favours production. † The interval includes zero under the two-way venue × date bootstrap (exact −0.05 to 0.58; within 10 −0.05 to 0.69).

Production minus	Within one level	Exact level	Within 10	MAE
(b) Hour-old reading, else curve + offset	−0.58 (−0.79 to −0.30)	+0.24 (0.03 to 0.51)†	+0.30 (0.02 to 0.65)†	+0.32 (0.15 to 0.44)
(a) Reading up to 12 h old, else popular times	+4.1 (3.6 to 4.7)	+1.7 (1.3 to 2.2)	+0.1 (−0.1 to 0.4)	−2.05 (−2.31 to −1.85)
Weekly curve alone (popular times)	+13.6 (12.3 to 14.7)	+24.7 (23.1 to 25.8)	+24.3 (22.3 to 25.7)	−8.40 (−9.02 to −7.70)
Curve + offset, no nowcast	+7.9 (6.9 to 8.7)	+20.5 (18.9 to 21.7)	+24.1 (22.0 to 25.5)	−6.54 (−6.96 to −6.05)
Rule engine	+17.8 (16.7 to 18.6)	+32.8 (31.3 to 33.9)	+35.6 (34.0 to 36.8)	−11.63 (−12.08 to −11.02)
Before: v2.6.0 as served	+20.0 (18.7 to 21.5)	+24.1 (22.6 to 25.4)	+28.3 (26.5 to 29.6)	−12.93 (−13.72 to −12.14)
Constant "Not Busy"	+3.1 (1.1 to 5.1)	+40.1 (39.0 to 41.0)	+30.7 (28.9 to 32.0)	−10.37 (−11.33 to −9.45)

Against popular times, curve plus offset, the rule engine, the configuration it replaced and the constant answer, every interval in Table 3 excludes zero, and the two-way bootstrap changes none of these conclusions. **H1 is supported.**

4.2 Where the accuracy comes from

Where the venue was read in the hour before, production serves that reading at full weight, the weight its fit chose; these "hour-old" readings are 41 to 77 minutes apart (10th to 90th percentile). Those rows, 48.3% of the window, carry 92% of the exact-level gain over popular times and 96% of the MAE gain. On a quarter of all readings (24.7%) the venue's value was unchanged from the hour before, and the served number reproduces it exactly.

The persistence baselines confirm where the accuracy comes from. Rule (a), the latest reading or else popular times, reaches 53.2% exact. Rule (b), the hour-old reading or else curve plus offset, is level with production on the exact level and slightly ahead on within one level (by 0.6 points) and MAE (by 0.32). The difference is the two- and three-hour nowcast weights (Section 4.4), and production has served rule (b) since 13 October. For the 52% of readings with no hour-old reading, the offset is the correction: it lifts curve plus offset over popular times by 5.7 points within one level and 1.85 MAE across the window.

The gain holds where the value moved. On the 59,681 readings whose value differed from the venue's previous reading, production names the exact level 41.0% of the time against 30.0% for popular times, and on the 28,325 changed readings with a reading from the hour before, 52.9% against 29.1%.

4.3 Across the month

Production's lead over popular times holds in every period and on every weekday. By period it names the exact level 52.5% of the time from 9 to 25 September (39,000 readings, before the switches went live), 55.7% from 26 September to 2 October (53,287, live in production) and 56.7% from 3 to 8 October (27,652), after the provider began returning fewer live readings, against 29.2% to 32.2% for popular times on the same readings. By weekday it runs from 54.4% to 56.7%, Friday and Saturday included, against 28.0% to 32.5%. Part of the rise across periods reflects more venues having an hour-old reading as the sweeps sped up (35.3%, 55.5% and 52.6% of readings); at the month's average mix the three periods read 59.0%, 53.0% and 54.8%.

Miami is read at two fixed hours a day, so no reading there is ever an hour old and its served number is curve plus offset alone. The nowcast's gain is a Pennsylvania result, measured on 112,728 readings.

4.4 What a reading is worth, by its age

The served number's accuracy is set largely by how recently the venue was read (Figure 4). Over the 30-day window, 48.3% of readings (57,900 of 119,939) had a reading from the hour before. On those, the served number lands within 15 points 83.0% of the time, within one level 88.6%, names the exact level 77.0% of the time and has an MAE of 9.99.

At two hours within-15 falls to 40.7% (14,778 readings) and at three to 38.6% (7,794), level with curve and offset on the same rows (39.7% and 38.7%). With the latest reading four to twelve hours old it is 43.9% (12,285), and with no reading in the last 12 hours it is 44.4% (27,182). On the three-day test the two-hour figure was 62.8%; over a month nearly all of a reading's value sits in its first hour. The card had published that 62.8% as its confidence for a two-hour reading, and on 8 October it was replaced with the month's figures.

The month also refined the weights. On the 14,778 readings whose latest reading was two hours old, curve plus offset alone beat the 0.85 blend by 2.73 MAE (interval 1.41 to 3.63) and 4.5 points within one level (2.4 to 6.1); the blend had been fitted on 1 to 5 September, when consecutive readings were identical four times in five. At three hours the 0.25 weight made no measurable difference. **H3 is supported:** the decay in accuracy follows the decay in correlation measured in Section 3.1, and is steeper. On 13 October production's two- and three-hour weights were set to zero, so it now serves rule (b), and the retrain gate compares against that stronger incumbent.

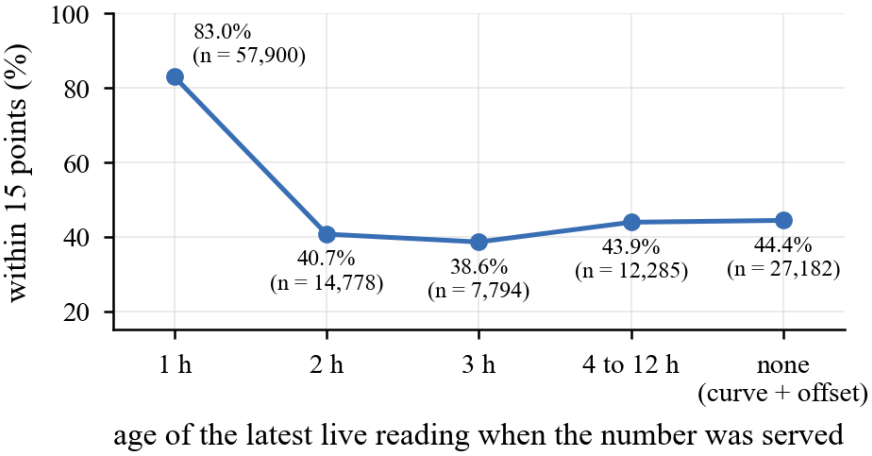


Figure 4. Accuracy by the age of the latest reading, over the 30-day window (119,939 readings). Since 13 October the card publishes the one-hour rate as its confidence when an hour-old reading entered and, otherwise, 42.7%: the within-15 rate of curve and offset alone over all 119,939 readings.

4.5 How far ahead

The served number is a nowcast, and its headline results are one hour ahead. To measure longer horizons, each of the 119,939 readings was forecast again as if from h hours earlier, using only the readings that existed then (Table 4). Two or more hours ahead the latest reading is too old to use, and production serves the venue's curve corrected by its offset. That number stays ahead of popular times at every horizon to a day: by 3.0 to 3.4 points of exact level, 4.7 to 4.9 points within one level and 1.2 to 1.4 points of MAE, with date-block intervals that exclude zero at every horizon. Within 10 is the exception (0.6 to 0.8 points lower), the lattice effect of Section 3.1. The large gains are one-hour gains; the smaller, steady lead is what a group checking the card hours in advance receives.

Table 4. Accuracy by forecast horizon on the same 119,939 readings, with the production rule served since 13 October: the hour-old reading at full weight when the forecast is made one hour ahead, and curve plus offset otherwise, using only readings that existed h hours before the scored hour. Persistence is the newest reading that existed then, at any age.

Hours ahead	Served exact ↑	Served within one ↑	Served MAE ↓	Popular times exact	Popular times MAE	Persistence exact
1	**54.7%**	**79.0%**	**17.68**	30.2%	26.40	53.2%
2	33.6%	69.7%	25.02	30.2%	26.40	32.3%
3	33.5%	69.6%	25.09	30.2%	26.40	30.3%
6	33.5%	69.6%	25.13	30.2%	26.40	28.4%
12	33.4%	69.5%	25.15	30.2%	26.40	28.5%
24	33.2%	69.5%	25.18	30.2%	26.40	33.1%

4.6 Where the gain comes from

The comparator rows of Table 2 separate the two corrections. Smoothing the curve and adding the offset together take the popular-times row from 64.8% to 70.5% within one level, from 30.2% to 34.4% exact and from 26.40 to 24.55 MAE, and move the bias from +4.89 to −1.45. Within-10 ends where it began (30.9% to 31.2%): smoothing moves predictions off the 5-point grid and lowers it to 26.4%, and the offset lifts it back, the grid penalty in miniature. The nowcast then does most of the work: within-10 rises from 31.2% to 55.2%, and the exact level from 34.4% to 54.9%.

The obvious remedy for the curve's poor calibration is to shrink it toward the mean. On the 9,903 paired rows, every correction that narrows the published spread buys MAE and spends within-10, so the system uses the per-venue intercept, which improves both on those rows (an in-sample estimate, since it is fitted and scored on the same readings). A leave-one-out test over 688 venues (11,804 held-out readings) repeats the pattern out of sample: the intercept buys 1.56 MAE against the bare curve (25.97 to 24.41) at a cost of 0.1 point of within-10, while a per-venue slope buys a further 0.99 MAE and costs 3.19 points of within-10.

The display map, part of the configuration production replaced, had bought 8.4 points of within-10 for 3.15 MAE on the 46,101 later held-out-city rows it was not fitted on. September's live readings reversed that verdict: the map cost 10.5 points of within one level (57.4% with it against 67.9% without, on 7,920 readings).

Volume against depth. A paired test scored two models on the same 17,322 later rows. One was trained with the scored venues' earlier history, 304,000 more rows in total; the other had never seen those venues. The model with the extra history scored 27.63 MAE against 27.14, worse by 0.49, in a single paired run. A two-head variant makes the same point from the other side. Trained on the unknown-provenance pool, it beats the shipped model by 0.88 MAE (95% interval 0.51 to 1.27). Held to the 4,223 verified labels available at the time, it loses by 1.59 MAE (interval 0.86 to 2.45). The limit is the supply of verified labels per venue, and live readings are the input that moves it.

4.7 The delta model on cities it never saw

The delta model is scored on the *gate slice*: the 67,249 realtime rows with a usable curve in Miami, Tokyo and Barcelona, the three cities it never trained on (Table 5). No held-out venue has earlier readings, so the offset is 0 there. These three cities have been the project's holdout since March: every version from v2.2 on was gated on them, and the score chain's clamp and one-point push were fitted on the first 30% of their dates, so they serve as its validation set. The pre-registered retrain adds a time holdout.

Table 5. Point accuracy on the 67,249-row gate slice, scored as the v2.6.0 ship gate scored it: the predicted delta clamped at ±30 and not rounded, with the offset at 0 (display map off).

Predictor	MAE ↓	R^2 ↑	Within 10 ↑
Weekly curve alone	31.48	−0.075	19.2%
Incumbent v2.5.0	30.77	−0.043	19.3%
Delta model v2.6.0	**29.42**	**+0.040**	**20.7%**

The model improves on the curve by 2.06 MAE (6.5%) and 0.11 R^2, and on the incumbent by 1.35 MAE. R^2 crosses zero: this is the first version to predict these cities better than the mean of the labels, a best-of-several result on the project's validation set. The gain holds in every held-out city (Figure 5): 2.26 MAE in Miami, 2.21 in Barcelona and 1.72 in Tokyo. On the five-level scale, with the delta clamped at ±50 and not rounded and a score of exactly 20 counted as *Not Busy*, it gains 2.6 points of exact level and 4.3 within one level, with 0.086 fewer levels of error on average (production counts 20 as *Quiet*; on that ladder the gains are 2.5, 4.4 and 0.089). Scored exactly as the score chain writes it, with the ±50 clamp and rounding, it reaches MAE 29.35, R^2 +0.042 and within-10 21.7%.

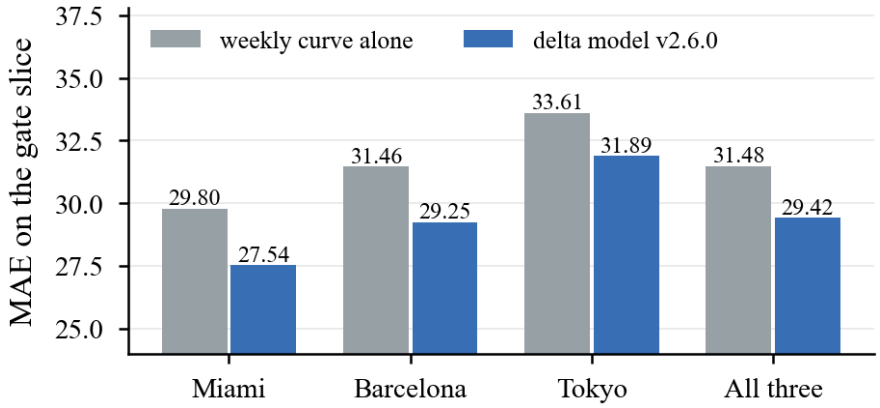


Figure 5. Delta model against the weekly curve on each held-out city of the gate slice.

A paired two-way cluster bootstrap (2,000 resamples, stratified by city) that resamples venues and city × month × weekday blocks puts the MAE margin over the curve at −2.06 (−2.75 to −1.44), the within-10 margin at +1.51 (0.44 to 2.65), the exact-level margin at +2.61 (1.55 to 3.79) and the within-one-level margin at +4.31 (2.93 to 5.80). Every margin excludes zero overall and in every city except within-10 in Miami and in Tokyo taken alone, so the claim for H2 rests on MAE and the level metrics. With the features, target and score chain held fixed and only the learner changed, in a rotation across the three held-out cities (each learner trained on two and scored on the third), the gradient-boosted ensembles did best: LightGBM had the lower MAE and XGBoost matched it on within-10.

Placed against prior work, the 6.5% comes with no recent history, where BysGNN's larger reductions read the previous 24 hours at every venue [2]. Its R^2 of +0.040 is positive at a granularity where the nearest peer-reviewed occupancy study's section-level forecasts a day or more ahead have R^2 near zero and often below it [14]. **H2 is supported on the held-out cities.** The model learned from spring rows of unknown provenance, and on verified live readings the online correction gives the stronger number, which is why production serves it.

4.8 What coverage is worth

A venue with no curve fell to the rule engine until 14 October. Since then it gets the typical curve of its category, refined by price and rating, which beat the rule engine on every scored slice: on 75,715 Philadelphia readings at 557 venues with each venue's own curve withheld, MAE 26.78 against 29.84 (−3.06, date-block interval −3.42 to −2.73) and the exact level 28.7% against 21.1% (+7.6, 6.9 to 8.6), with

the same direction in the Lehigh Valley and Miami. The comparison below predates that switch. Given identical inputs on 5,025 verified live readings in Pennsylvania at venues that hold their own curve, with the events lookup disabled for both, the model path beat the rule engine by 1.82 MAE and 11.4 points of within-10 (11.2 points grid-equalised). The bare curve beat the model path on MAE by 1.61 while losing on within-10: on these bimodal labels the two metrics reward different predictions.

4.9 The first test

The online correction was shipped on the result of a first, three-day test, 4,183 readings from 2026-09-06 to 09-08 (Lehigh Valley 3,615, Miami 568), and went live on 26 September. On those readings the served number named the exact level 56.0% of the time and the constant answer tied it on within one level, 78.8% each (production's three-day row is the last row of Table 2). The 30-day test confirms that result on 29 times as many readings, and it is the measurement this paper rests on.

4.10 The findings on public real counts

Every figure above is measured against the provider's estimates. To see how counted crowds behave, I repeated the analyses on 15 datasets of real counts, under an analysis plan whose pass rules were fixed and hashed before any count was downloaded (two amendments, each hashed before its data were fetched, added datasets and horizons): hourly pedestrian counters in Brisbane, Melbourne (2019 and 2025 to 2026), Dublin, Cork, Dún Laoghaire, Leeds, Auckland, Basel and Zurich; trail counters in Cambridge, Massachusetts and Seattle; hourly entries at 61 New York subway stations; and people inside a university gym and six reading areas of the KIT Library in Karlsruhe, used with the library's permission. Together they hold 5,321,507 hourly counts at 315 sites in seven countries. Each site's counts were rescaled to 0 to 100 so that errors read on the paper's scale, typical-week curves were built from the eight weeks before each scored day, and intervals are week-block bootstraps (Table 6, Figure 6).

Table 6. The provider's live readings beside public real counts, one hour ahead. Real counts are pooled over 15 datasets with equal weight: 13 count people passing a sensor and two count people inside (a gym and a library), shown on their own as well. Each source's MAE is on its own 0 to 100 scale, so predictors compare within a column. The provider's newest reading is the latest at any age; on counts every hour has one.

Measure	Provider's live readings	Real counts, 15 datasets	People inside: gym and library
Weekly curve as a forecast: slope, r	0.50, 0.36	0.92, 0.79	0.76, 0.62
Quiet hours, busy hours: observed minus curve	+11.5, −13.9	+0.3, −1.4	−0.5, −4.0
Departure correlation at 1, 2, 3 hours	0.74, 0.26, 0.16	0.74, 0.61, 0.54	0.83, 0.78, 0.73
MAE: weekly curve	26.40	9.58	14.36
MAE: newest reading as the forecast	21.94	10.36	8.23
MAE: curve + trailing offset	24.55	7.75	7.85
MAE: curve + latest departure		**6.41**	**6.12**

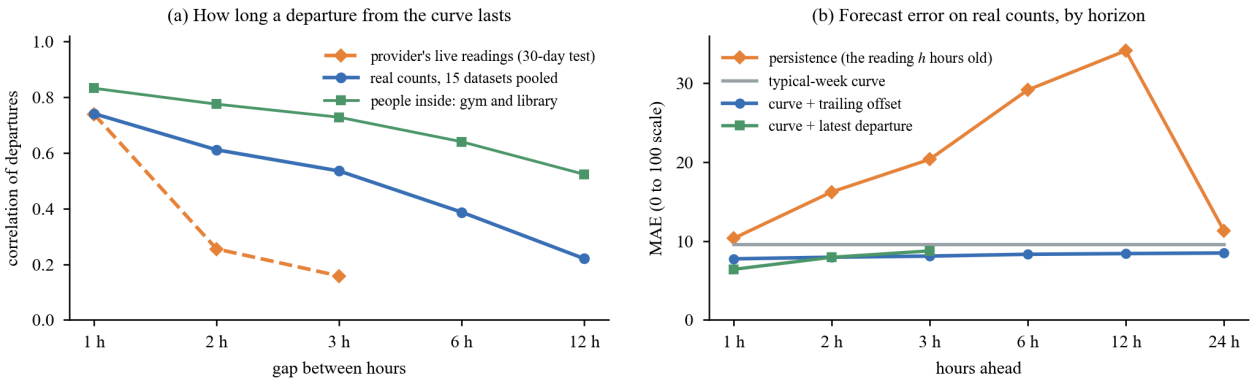


Figure 6. The provider's live readings beside 15 real-count datasets. (a) How long a venue's departure from its typical-week curve persists. (b) Forecast error on the real counts by how far ahead the forecast is made (pooled, active hours).

Part of the result carries over. A site's latest departure from its curve is its most valuable one-hour input, and the curve alone is never the best forecast: adding the latest reading's departure to the next hour's curve gives the lowest one-hour error in 14 of the 15 datasets, 33% below the curve alone (MAE 6.41 against 9.58; this predictor was registered without a pass rule), and the trailing offset improves on the curve at every horizon to 24 hours.

The provider-specific findings do not carry over as stated, and the way they fail points to the provider's readings and curves, though the comparison also differs in what is counted, how curves are built and the labels' lattice. Real counts follow their own typical-week curve closely (slope 0.92, r 0.79), so the curve is a strong forecast, and the raw hour-old count loses to it in 9 of the 15 datasets; under the registered pass rules the first finding replicated in 4 datasets and the calibration and decay findings in none. Real departures also last longer, with a correlation of 0.61 two hours on and 0.54 three hours on, where the provider's falls to 0.26 and 0.16. The provider's live value therefore swings around its curve more widely and settles back faster than counted crowds do, while repeating itself from one hour to the next. On its readings the latest value used as a level is the best one-hour forecast; on real counts the latest departure carried onto the next hour's curve is, the variant production tested and set aside on the provider's sticky readings.

The two datasets that count people inside a place, as venue busyness does, side with the provider's readings on the first finding. At the gym and the library the hour-old count beats the typical-week curve by a wide margin (exact level 66.0% against 48.1%, MAE 8.23 against 14.36), the curve fits less well (slope 0.76, r 0.62) and departures persist longest (0.73 at three hours). Where the quantity is people inside, the latest reading is worth even more than on footfall, and carrying its departure onto the curve is better still (MAE 6.12). For a real-count source such as Flux, the evidence favours exactly that.

Every other count I could use. Three more checks, run after the registered analysis and outside its pass rules, use the rest of the data. The KIT libraries that close at night miss the registered rule only because Seatfinder logs nothing while they are closed; six of them have enough history to score at their open hours (12,248 hours), and they repeat the library result. The hour-old count beats the typical-week curve (MAE 8.13 against 15.32, exact level 68.3% against 47.0%), departures persist (0.93 at one hour, 0.80 at three), and carrying the latest departure onto the curve is best again (MAE 5.32, exact level 78.6%).

Five rooms at the National University of Singapore (ROBOD) agree on both of their measures. On the dataset's own occupant counts (1,046 scored hours, up to 38 people in a room) the hour-old count scores MAE 12.18 (10.66 to 13.78) against 21.63 (18.96 to 24.53) for a two-week curve, and on connected Wi-Fi devices (2,056 hours) 5.53 (4.92 to 6.15) against 11.77. So do the 153 usable indoor Wi-Fi access points across the 24 buildings of the Universitat Politècnica de Catalunya's Campus Nord in Barcelona (71,926 scored hours, April to June 2023): the hour-old count beats the curve at 152 of the 153, MAE 8.87 (8.66 to 9.08) against 16.60 (16.31 to 16.94). Finally, to test whether the provider's readings follow real people at all, I set the provider's typical week for 695 New York venues beside the hourly entries at the subway station nearest each one: 34 stations with at least five venues within 400 m, March to May 2026. The two weekly rhythms match closely (median r 0.74, and 0.81 for the city as a whole), and at 28 of the 34 stations the provider's venues match their own station better than they match other stations (mean gain in r 0.033, interval 0.017 to 0.050), so the provider's curve carries each block's own rhythm, not only the city's. Station entries are not the crowd inside a venue, so this checks the provider's rhythm rather than its level.

5 Discussion and the pre-registered retrain

5.1 What it means

The latest reading is the lever. Adding the venue's latest live reading raises within-10 from 31.2% to 55.2% and the exact level from 34.4% to 54.9% over 30 days; with the offset, the served number cuts the curve's MAE by 31.8% against the provider's live readings. The gain is concentrated in the latest reading: where an hour-old reading exists, using it at full weight is the strongest rule this project found, and the corrected curve carries the other half of venue-hours. For anyone building on typical-week curves, the lesson is to score every forecast against the latest observation as well as the curve [6]: on the provider's readings the latest value wins as a level, and on real counts the latest departure carried onto the curve wins (Section 4.10).

The model supplies a correction. On the gate rows its output correlates with the raw curve at 0.9638: the curve supplies the rhythm and most of the level. A 6.5% MAE reduction is what the spring data supports without any observation of the venue's recent state.

Evidence density is the binding constraint. The spring corpus holds about 25 realtime rows per venue across 168 weekly slots, and, in one paired run, 304,000 more rows of the same kind did not help. On a bimodal target, MAE and within-10 can give opposite instructions, as the curve corrections show; a live reading moved both at once, cutting MAE from 24.55 to 18.00 and raising within-10 from 31.2% to 55.2%. The hourly live collection is the depth the problem asks for, and a reading's value lasts about an hour, so how often a venue is read matters more than how long its history is.

Who it helps. For a group choosing where to go, moving the exact level from 30.8% to 54.9% turns the card into an answer that is right more often than not, measured against the provider's readings. For a venue, a number people trust about the present can make a quiet hour worth visiting and move some demand off the peak. For anyone building on typical-week curves, the measurements carry over without any of this code: judge such a curve as a forecast and against persistence, read a within-k

metric on lattice-valued labels beside its on-grid rate, and prefer fresher readings of each venue to more rows of the same kind. Flock receives only venue-level scores from the busyness provider, which builds them from aggregated, anonymised signals; no individual's data reaches Flock.

Scope. The labels are the provider's estimates rather than counts, and the test covers three of the 34 cities in one season; Section 15 gives the details.

5.2 The pre-registered retrain

This paper is a snapshot taken before the early-November retrain. That retrain will take the five live features of Table 1 as model inputs, train with live readings carrying 80% of the loss, drop the month family and the row-type flag, and add the sports-schedule features, built from 627 professional and college games refreshed daily. It will be gated against production as served at gate time, which since 13 October is rule (b), and shipped only if it wins; the configuration tested in this paper and the constant answer will be reported beside the verdict. The run was set for about 19 to 20 October and has moved to early November, after Halloween, so that, if the provider's feed returns, the held-out window moves past the 30-day test; I will update this paper with its held-out results. That update adds the strongest test in this project: a model trained on a month of verified live readings, scored on a 14-day window whose decision rule was fixed before its data existed.

The rehearsal. Scored on the first test's 4,183 readings, a model trained with those five live features on 3,579 earlier live readings reached 79.7% within one level, +21.2 points over v2.6.0 as served (95% interval 15.8 to 24.2). The live features carry 47% of its split gain and are worth +10.0 points of within-10 over the same candidate without them. It still trailed production by 12.2 points of within-10 (interval −16.3 to −2.7) and by 1.9 MAE. The band gate refused it on the MAE guard and the three-date sample, and I added the within-10 criterion so that a loss like this one is refused on its own.

A pre-registered test. Hypotheses H1 to H3 were stated after the analyses that test them (see the appendix). The pre-registered retrain is the first test in this project whose hypotheses, data split and decision rule were fixed before its data existed. They are fixed in code: the band gate's thresholds and the 14-day time holdout were committed on 2026-09-25 and 2026-09-26, before any held-out reading existed. One element was set later: on 13 October, after the month's analysis on dates that overlap a possible held-out window, the incumbent it is compared against was simplified to rule (b), which makes the comparison harder for the retrain.

- **H4.** The retrain scores higher than production as served on within one level over the held-out 14 days, and its date-block 95% lower bound is above −1.0 point.
- **H5.** It is not worse on within 10, by the same bound.
- **H6.** It beats the weekly curve on within one level with a lower bound above zero, and its band MAE is no worse than the curve's.
- **Guards.** At least 1,000 readings over at least 5 dates; production never saw the window; no city with 300 or more readings falls more than 2 points behind production; within one level is not below the rule engine's; MAE rises by at most 1 point.
- **Analysis.** 2,000 date-block bootstrap resamples. The constant-answer reference is recorded beside every verdict.

- **Decision.** The retrain ships only if all of these hold. Otherwise production keeps serving, and the result is reported either way.

The sample after the provider's drop. On 25 September, at about 2,100 readings a day, I projected about 90,000 to 95,000 provenance-verified live readings by mid-October, with the last 14 days held out. From 25 September the sweeps ran in parallel, and from 26 September to 2 October the collection gathered about 7,600 model-served readings a day. From about 23:00 on 3 October the provider answered the same hourly calls with fewer live readings, about 4,600 a day from 3 to 8 October, 61% of the week before, and since about 16:00 on 12 October it has returned none; the provider has been asked why. The registered rule anchors the 14-day window on the last date with live readings. If the feed does not return before the run, the window is 29 September to 12 October, which holds far more than the floor of 1,000 readings over 5 dates. Ten of those dates fall inside the month reported in this paper, so production's accuracy on them is already known. If the feed returns before the run, the window moves onto later dates and the overlap disappears. The retrain trains only on earlier dates, and none of its settings were chosen on these ten days.

Among the readings the provider did return, the drop left accuracy where it was. The share with a reading from the hour before moved from 55.5% to 52.6%, and production named the exact level 56.7% of the time after the drop against 55.7% before. Readings that were never returned cannot be scored. The decision rule does not change.

How strict the gate is. With about 60,000 held-out readings, an illustrative size, a 95% interval on a within-10 difference is about ±0.6 points if readings were independent, and ±1.3 to ±1.8 points with 5 to 10 times variance inflation, the clustering the first-test rehearsal measured, used here as an illustration (the 30-day window measured about 16 times on the exact-level margin over popular times). At that inflation a retrain that exactly ties production would see its lower bound fall below −1 point and be refused; to clear it, it would need a within-10 win of about 0.3 to 0.8 points. The gate scores paired date-block differences, whose interval depends on how the two systems disagree across readings and dates, so the real requirement may be lower. I have kept the bound strict on purpose: the rehearsal is the case it exists to stop.

6 Conclusion

From 26 September to 13 October the number Flock served corrected the venue's curve with its recent live offset and a nowcast of its latest reading. On 119,939 live readings its weights never saw (9 September to 8 October 2026) it named the exact crowd level 54.9% of the time, landed within one level 78.4% of the time (a constant answer reaches 75.2% but is exact only 14.8% of the time) and came within 10 points 55.2% of the time; the configuration it replaced scored 30.8%, 58.3% and 26.9%, and the served number's MAE is 31.8% below the popular-times baseline's, measured against the provider's live readings. The three-day result it was shipped on held up on 29 times as many readings, across all seven weekdays and every period measured. The gain comes from the venue's latest reading: with an hour-old reading, 77.0% of served numbers land on the exact level, and production now serves that reading at full weight with the corrected curve as its fallback. The freshest reading of a venue is worth more than any correction to its curve. Two to 24 hours ahead, when no fresh reading exists, the corrected curve

still names the exact level about 3 points more often than popular times. On 5.3 million public real counts the provider-specific findings do not hold as stated: counted crowds follow their curves far more closely than the provider's readings do, and the latest reading helps as a departure carried onto the next hour's curve, the lowest one-hour error in 14 of 15 datasets (an exploratory comparison); where the counts are people inside a place, the hour-old count beats the curve, as on the provider's readings. Separately, a gradient-boosted delta model, which production does not serve today, beats the curve on whole held-out cities without any recent-history input. The pre-registered retrain will test whether a model that learns from live readings can beat the served number, under a decision rule fixed before its data existed.

PART II

Counting the people who are there

Owl, an on-device thermal people counter for Flux, Flock's venue sensor.

7 Counting people on a thermal camera

Flux is a small venue sensor that a venue could install to count the people actually there. It is built around a Raspberry Pi 5 and a FLIR Lepton 3.5, a radiometric thermal camera that reports a temperature in degrees Celsius for each of its 160 × 120 pixels across a 57° horizontal field of view [15]. The Pi reads the camera through a PureThermal 3 board over USB. No Flux unit is installed at a venue yet, and Flock's crowd number does not read sensor data.

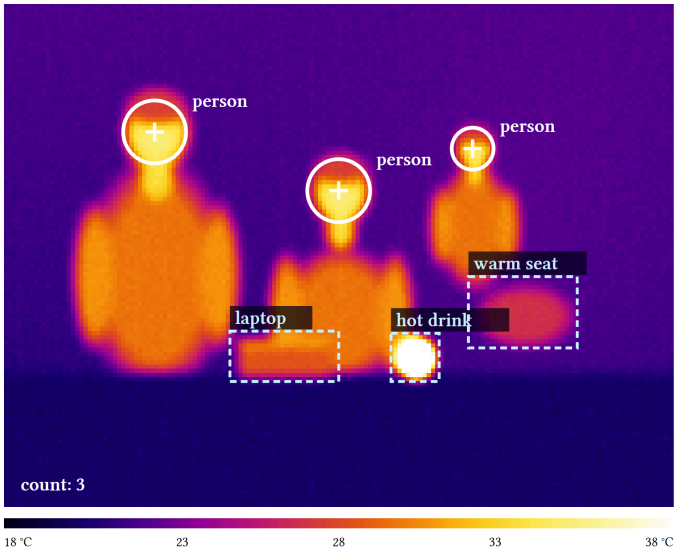
A thermal camera records heat instead of light. Through the Lepton's lens a head three metres away is about 8 pixels wide, and a hand 30 cm from the lens is about 45. At 160 × 120 thermal pixels Owl is built to count people, never to identify them. Flux reduces each frame to a count on the device and throws the frame away (Figure 7).

Thermal people counting is usually studied with low-resolution array sensors on the ceiling, such as the 32 × 24 MLX90640 in PUT Thermo Presence [16], whose counting models have been benchmarked on resource-constrained hardware [17], and with higher-resolution cameras for segmentation and pose [18, 19]. Flux needed a counter that runs on its own camera, on the sensor itself.

The first counter on Flux was a heat-threshold rule. It bins the frame 4 × 4, estimates the room as the 20th percentile of the frame, learns fixed warm spots such as radiators per cell over time, keeps cells warm enough to be skin, groups them by eight-connectivity, and counts groups of at least 6 cells, about one head at three metres. Its settings were tuned on a bench with a real person and against constructed scenes, and it works for well-separated people at moderate range.

The rule fails in predictable ways. A hand held up to the lens counts as a person, and spread fingers count as two. A mug of coffee or a laptop can count as somebody. A packed room merges into one warm mass. Telling these apart is a question of shape, which is what a learned model does well.

Owl is the model that replaced the rule. It is a 718,550-parameter CenterNet-style network [20] that places a point on every person and a named box on nine kinds of warm objects. It runs in 14 to 16 ms per frame on one core of a Raspberry Pi 5, with no GPU and no round trip to a server, and it keeps nothing from one frame to the next. I recorded no footage of people to train it. Owl learns from a generator of thermal scenes built on measured skin and clothing physics, and from six real thermal datasets used under commercial licences or with their authors' written permission.



Camera	FLIR Lepton 3.5, 160 × 120, 57°
Input	degrees Celsius, every pixel
Model	718,550 parameters
On the sensor	14–16 ms a frame, 1 Pi 5 core
Names	person + 9 kinds of warm thing
Counts	people only
Leaves the device	counts and a sound level
Footage I recorded	none

Figure 7. What Owl reads and what it reports. An illustrative 160 × 120 frame, drawn the way Owl's training generator draws a scene, shown at the camera's true resolution in the iron palette the Lepton uses. Owl puts a point on each person's head and a named box on everything else warm. Only the points are counted: this frame counts as three.

8 How Owl works

Each training frame is labelled with one point per visible person, at the centre of the head, or at the centre of the visible body when the head is out of view. Every other warm object gets a box and one of nine kinds: hand, pet, hot drink, food, laptop, screen, heater, lamp and warm seat. Counting people then reduces to counting the peaks in the person map, and that count is exactly the quantity a venue needs.

The other nine maps let the sensor's screen show why something was not counted, and they give the network somewhere to put the things it must not count.

The network. Owl sees two input channels. The first is absolute, (t − 30) / 8, so a 60 °C mug can never look like skin. The second is relative to the frame's median, (t − median) / 4, so the several degrees an uncalibrated Lepton reads high or low do not move a person out of range.

An encoder of residual blocks [21] halves the resolution four times, with dilation at the deepest stage [22, 23] to widen the view: a hand filling half the frame must be seen as one hand and not five warm blobs. A top-down path fuses the deep features back to a 40 × 30 grid. Three heads read that grid: ten heat maps, a box head that predicts each object's reach, and, for the crowd model, a density head. The main model exports to ONNX as a 2.9 MB file.

Generated frames. Owl's first version learned only from generated frames, and the generator stayed in every training batch after that, at 40% of frames for the shipped model. It draws a room at some temperature, people at every distance and pose, and the objects that fool a threshold, at twice the camera's resolution. It then averages down, blurs for the lens, and adds the camera's noise, fixed-pattern column stripes, dead pixels, vignetting, gain error and a crooked mount.

The temperatures come from published thermography of skin and clothing. A forehead reads 33.4 °C in a 22 °C room, plus 0.14 °C for every degree of room temperature. Clothing sits a garment-specific share of the way from the room to the skin, so a winter coat reads barely above the room. A seat someone just left is 4 to 8 °C above the room, and 20% of frames are drawn in a hot room, at 29 to 34 °C. This measured model replaced hand-picked temperatures after the first round of real-footage tests.

Scenes include standing, sitting and lying people, children, hats, hoods, masks, bald heads, bare torsos, winter coats, people hidden behind tables and bars, and one person partly behind another. They also include things that must not be counted: hands at the lens, reflections of people in glass and shiny floors, warm footprints, sun on the floor, candles and bulbs hotter than the camera's range. The idea is domain randomization [24]: vary everything the model should ignore so that the real world looks like one more variation.

Licensed real data. Six real thermal datasets and one set of rendered frames entered training over the project (Table 7). Two are from the Lepton 3.5 itself. Sources that store temperatures are used in degrees Celsius. Sources stored as 8-bit images carry no temperatures, so each frame is mapped onto a plausible room-to-skin range drawn afresh every time it is used (low end 16 to 28 °C, span 8 to 16 °C, gamma 0.8 to 1.25), which stops the network from treating any single mapping as truth. IISER's frames are colour images in the camera's iron palette, and I convert them back to one channel by luminance, which rises monotonically along that palette.

Real datasets label people and nothing else, so for real frames the loss on the other nine maps is switched off. AAU, SenSys and SDT mark bodies rather than heads. For those, the head point is estimated from the box and its target is spread over the pixels the head could occupy (σ = 5 px).

Table 7. Training data for the shipped model. Frames are training-split frames. The real and rendered sources together give 38,648 training frames and 70,402 labelled people, 60% of what the model trains on.

Dataset	Camera and view	Frames	Role
Generated scenes	Physics-based scenes through the Lepton's optics and noise	generated on the fly	Every person and nine kinds of object labelled; 40% of training frames
PUT Thermo Presence [16]	MLX90640 32 × 24, ceiling	8,472	First real frames mixed in, with OpenThermalPose2 (owl-3)
OpenThermalPose2 [19]	Side view, indoor and outdoor	8,846	Side views of people (owl-3)
AAU VAP Trimodal [18]	Wall-mounted, meeting rooms	3,964	Meeting rooms (owl-4)
SenSys 2021 set [25]	32 × 24 rooms with hot laptops	1,239	Warm laptops beside people (owl-4)
SDT, TU Wien [26]	Lepton 3.5, elevated indoor, one person or none	4,800	Flux's camera model (owl-4.5)
SDT rendered frames [26]	Rendered to the Lepton's look	8,000	Ended the trade between Flux's camera and the meeting rooms (owl-4.8)
Thermal Vision, IISER Bhopal [27]	Lepton 3.5, classroom of 12 to 15 and lab of up to 6 (up to 10 and 5 labelled faces)	3,327	Real crowds on Flux's camera model (owl-4.9)

Training. The loss is CenterNet's penalty-reduced focal loss on the heat maps [20, 28], with $\alpha = 2$ and $\beta = 4$ as in CornerNet [29], masked by which maps a frame actually labels, an L1 loss on the log of each box's reach, and, for the crowd model, squared error per density cell plus a tenth of the error in the total. I train with AdamW and a one-cycle learning-rate schedule in batches of 64. Eight CPU workers generate frames on the fly for an RTX 5080, which trains at 1,000 to 1,140 frames a second. Later versions start from the previous version, keeping every layer whose shape matches. Over four days the project ran 19 full training runs plus five short trials, 581,700 steps in all, which is 37.2 million training frames.

Decoding on the sensor. The sensor runs the ONNX file with ONNX Runtime [11] in full precision on one core. A person is a cell at least as confident as all eight neighbours and above 0.4. Rarer kinds must clear a higher bar before they are named, from 1.25 times the person threshold for hands and hot drinks to 1.75 times for pets, because they are learned from fewer examples. A person peak is dropped when a screen, laptop, heater, lamp or warm seat is more confident in the same 3×3 neighbourhood.

Two physical checks then run on a 9×9 patch around each person. Its warmest pixel must be at least 1 °C above the room, taken as the frame's 10th percentile, and its median must not exceed 45 °C, which no person reaches.

The crowd model hand-over. Peak finding merges two touching heads into one. Density-based counting [23, 30, 31] regresses a map whose sum is the count, and it degrades gracefully when people overlap. Owl uses both. When the main model finds five or more people, a second crowd model with a density head reads the same frame, and the published count is the highest of the main model's peaks, the crowd model's peaks and the rounded density total.

9 How it was tested

Test splits are the datasets' own, or, where a dataset has none, contiguous ranges in time: SDT by frame range 60/15/25, and IISER by time within each room 60/15/25 with ten frames dropped at every boundary, because neighbouring frames are near copies.

Checkpoints, benchmarks and a sealed test. Each run keeps the checkpoint with the lowest combined error on held-out generated and real validation frames, never the lowest training loss. Every version was then graded on the six test splits, which served as the development benchmarks for choosing versions, data shares, decoding settings and the crowd hand-over point. No test frame was ever trained on: the training cache holds training and validation frames only, with no exact duplicates of a test frame. A sealed test, held out from all of these choices and scored once, is reported with the results.

Six real benchmark sets. Accuracy is measured on six test splits that no model trained on:

- the Lepton 3.5, Flux's camera model, in indoor rooms (SDT, 2,000 frames);
- the Lepton 3.5 in a crowded classroom and lab (IISER, 1,375 frames, 4,882 labelled people);
- a ceiling camera (PUT, 2,347 frames, 8,043 people);
- rooms with hot laptops (SenSys 2021, 265 frames);
- wall-mounted meeting rooms (AAU, 813 frames, 1,061 people);

- a side view (OpenThermalPose2, 2,151 frames, 3,872 people).

A frame is *exact* when the count equals the truth and *within one* when it is off by at most one person. The rule is scored on the same frames, as deployed and, separately, with its settings tuned on each set's validation split. Intervals are block bootstraps that resample runs of consecutive frames within a scene or sequence (2,000 resamples), because neighbouring frames are near copies.

Two properties of the sets are worth knowing. IISER's labels mark visible faces, so a person facing away from the camera is absent from its truth; a recount is reported below. And some splits share scenes: AAU's three meeting rooms appear in training and test, and one PUT test sequence continues a training sequence. Removing that sequence and every test frame nearly identical to a training frame raises Owl's PUT score from 92.9% to 96.0% exact and moves no other set by more than 0.1 points: near-copies do not inflate the results. AAU's shared rooms were not re-split, so its figure is an in-scene result, and IISER's test frames share their two rooms with training.

The stress test. The second measure is 41 hard situations, each generated on purpose 150 times from a seed fixed per case and scored alone, counted exactly as the sensor counts, crowd model included. Every real set is also split by situation (by pose for SDT and by number of people elsewhere), and anything under 90% exact or 97% within one is flagged as a blind spot.

The stress cases come from the same generator as 40% of the training data and guided which cases training emphasised, so the stress test checks coverage of known hard cases; the real sets carry the headline numbers.

The crowd model's hand-over point was also chosen on test frames. Handing over from three, four or five people gave 36.9%, 37.5% and 39.3% exact on the classroom crowds, and five also gave the best stress mean, so the sensor keeps five.

10 Results

The shipped pair is owl-4.9 and its crowd model. On five of the six real benchmark sets it counts 82 to 96% of frames exactly, where the rule as deployed manages 9 to 77%, and it is within one person on 96 to 100% of frames. On dense classroom crowds filmed with the Lepton 3.5, the hardest set in this study, it is exact on 39% of frames and within one on 74%, against the deployed rule's 33% exact (Table 8, Figure 8); on this set the two are statistically close on exact counts, and Owl's lead grows with crowd size.

Table 8. owl-4.9 with its crowd model against the heat-threshold rule, scored on identical frames, with 95% block-bootstrap intervals for Owl's exact rate (Wilson intervals for the sealed tests and the stress mean). Percent of frames. The last column is the rule with its bin, margin and minimum cluster size tuned separately on each set's validation split; the sealed sets have no validation split.

Benchmark set	Owl exact	Owl within one	Rule exact, as deployed	Rule exact, tuned per set
Lepton 3.5, indoor rooms (SDT)	89.7 (85.4–93.4)	100	77.1	93.7
Lepton 3.5, classroom crowds (IISER)	39.3 (31.2–47.7)	73.9	33.0	40.2
Ceiling camera (PUT)	92.9 (86.9–97.2)	99.7	9.2	29.4
Rooms with hot laptops (SenSys 2021)	95.5 (91.7–98.2)	100	71.3	82.6
Wall-mounted meeting rooms (AAU)	93.0 (86.4–98.0)	99.1	51.5	75.5
Side view (OpenThermalPose2)	82.1 (74.4–89.2)	96.3	36.6	60.6
Sealed test: office, 1–2 people (OpenThermalAction)	94.3 (91.1–96.4)	100	40.7	
Sealed test 2: another 32 × 24 sensor (SKKU)	33.7 (24.7–44.0)	86.5	79.8	
Sealed test 2: outdoor handheld camera, people and dogs	41.0 (33.8–48.6)	86.1	16.9	
41-case stress test, mean	94.7 (94.1–95.2)	99.3		

A tuned rule. As a stronger baseline, I also tuned the rule's settings separately on each set's validation split. Owl, one model for all six sets, leads the tuned rule by 13 to 64 points on four of them. On the two Lepton sets the tuned rule is level with or ahead of Owl: 93.7% against 89.7% on SDT, whose frames hold one person or none, and 40.2% against 39.3% exact (84.6% against 73.9% within one) on IISER. Adding the sensor's learned background mask made the tuned rule worse on every video set, because people who sit still fade into it. A threshold tuned to one camera is a strong baseline in simple scenes; Owl's advantage is a single model that holds across the six cameras and views it was trained on and ignores the warm objects that fool a threshold.

A sealed test. To measure Owl on footage that guided nothing, I set aside 300 frames from the office test split of OpenThermalAction, a public MIT-licensed thermal set that no Owl version, setting or threshold had seen, spread across 269 clips of one or two people, at least 30 frames apart within a clip. Owl's and the rule's counts were computed and stored first. The frames were then counted visually by two separate passes of the same vision-language model, each blind to both outputs and to the other, with a third pass wherever they disagreed; the passes agreed on 93.9% of the 734 frames counted this way, and on the sealed set the counts match the dataset's own design (single-person and two-person

clips) on 298 of 300 frames. Scored once, Owl counted 94.3% of the sealed frames exactly (95% interval 91.1 to 96.4) and was within one person on every frame. The deployed rule counted 40.7% exactly and 88.7% within one. Owl's margin is 53.7 points (clip-block bootstrap 47.3 to 60.0).

Beyond its training cameras. A second sealed test asked how far Owl travels. Owl's and the rule's counts were again stored before scoring, and truth is each set's own person boxes (duplicates merged). On 89 frames from a different 32 × 24 sensor (the SKKU MLX90640 set, MIT licence), Owl counted 33.7% exactly against the rule's 79.8%: it took close, blurred bodies for screens and dropped them. On 166 outdoor photos from a handheld Seek camera (Thermal Dogs and People, CC BY 4.0), Owl led the rule 41.0% to 16.9%, though on the 70 frames showing only a dog it still counted a person in 59%. A third public set, the only one with many frames of three or more people, was scored under the same protocol and then withdrawn because its licence could not be confirmed; Owl trailed the rule on it. A third sealed test used the one licence-clean public set found with indoor groups of three or more: 109 unseen frames from a 32 × 32 thermopile on a ceiling (Klir et al., GPL-3.0), one to five people in each. As on the other low-resolution sensor, the rule did better, 18.3% exact against Owl's 9.2%, and 24.7% against 3.9% on the 77 frames of three to five people: Owl undercounts the faint, blurred bodies of a sensor it never trained on. Owl's accuracy belongs to the kinds of camera and scene it was trained on, which is why Flux's own camera, in real rooms, is the next test.

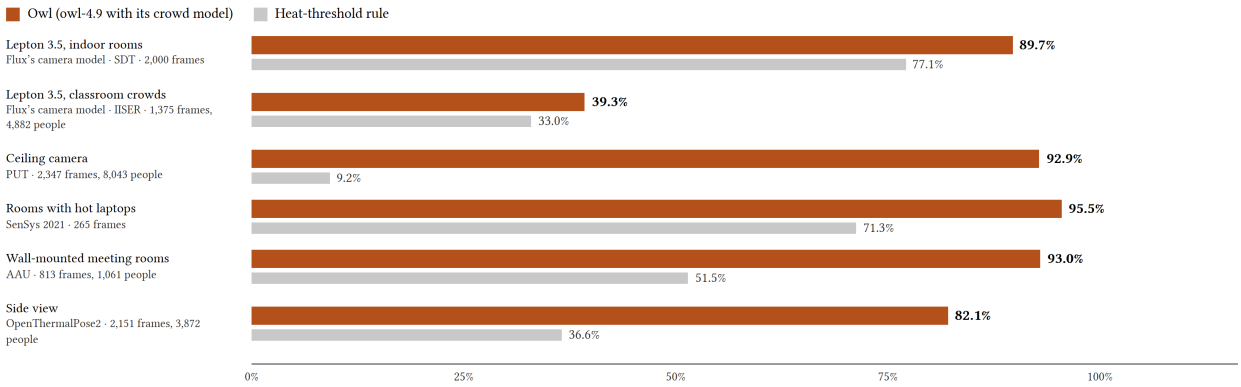


Figure 8. Frames counted exactly right on six real benchmark sets that no model trained on, owl-4.9 with its crowd model against the heat-threshold rule as deployed. The first two sets come from the FLIR Lepton 3.5, the camera model Flux uses. The rule is scored on identical frames; Table 8 adds the rule tuned per set.

The deployed rule's worst set is the ceiling camera, at 9.2% exact, where Owl reaches 92.9%. Its best is the Lepton in indoor rooms, at 77.1%, against Owl's 89.7% exact and 100% within one.

Crowds on the Lepton 3.5. The IISER classroom set is the only real crowd I have from the Lepton 3.5: rows of 12 to 15 students, up to ten with a visible face, each row partly hidden behind the one in front. owl-4.8 had never seen anything like it and overcounted badly, averaging 8.1 people on frames that held 3.6. Training the main model on the classroom frames lifted exact counts from 11.5% to 28.6%. Retraining the crowd model as well took them to 39.3%, and in rooms of six to ten people moved within-one from 3.4% to 96.1%. The average error fell from 4.66 people per frame to 1.03.

The new data did not cost the other sets. I re-graded owl-4.8 and owl-4.9 on the full test splits with identical code. On the five earlier sets no exact-count rate moved by more than 0.6 points and no within-one rate by more than 0.8.

By crowd size, Owl's strength is the crowd. On frames with six to ten labelled people it is within one 96.1% of the time, against the deployed rule's 4.5%. Sparse classroom frames are the opposite case: every classroom frame labelled empty or with one face falls in IISER's test split (training holds 17 two-face classroom frames and none with fewer), so Owl never trained on an empty or one-person classroom. I had these 334 frames and 100 others recounted visually, counting every person with any part in view, in the same blind two-pass procedure as the sealed test. The recount shows that the face labels undercount (frames labelled empty held 1.1 people on average) and narrows Owl's error on sparse frames. On frames holding one or two people the rule remains more accurate (85% exact, against Owl's 21% to 22%), in part because Owl puts points on warm seat backs; on frames holding three or more, Owl leads (43% to 44% exact, against 30% and 0%). With the recounted frames substituted, the IISER test reads 41.5% exact and 77.7% within one for Owl, against 40.4% and 69.2% for the rule. Real sparse rooms from Flux's own camera are the clearest next addition to its training data.

Hard cases. In generated scenes, every warm object that fools the rule by design is counted correctly on 98 to 100% of stress frames, where the deployed rule manages 14% to 85%: a hand at the lens, a hot mug, a monitor, a radiator, a pet, a reflection, warm footprints, a candle. Four cases fall under 90% exact: people lying flat (87.3%), a bare torso filling the view at the lens (78.0%), one person behind another (60.7%, but 99.3% within one) and a crowd of ten, the weakest, at 31.3% exact (95% interval 24.5 to 39.1) and 75.3% within one.

The real benchmark sets show the same pattern. A side view of three to five people is exact on 40% of frames (85.5% within one). On SDT, lying people are exact on 81% of frames and sitting people on 83%, though both are always within one.

A real room showed what no test set did. On the sensor in a real room, owl-3 named a pet in a room with none, sometimes counted a monitor, and missed a shirtless torso close to the camera. Three changes answered them: higher naming bars for rarer kinds, dropping a person where a screen-like object is more confident, and bare skin and bodies filling the view in the generated frames. On the stress test a shirtless torso at the sensor went from 16% under the first owl-3 decoder to 78% under owl-4.9.

Naming everything else. On 3,000 held-out generated frames, owl-4.9 finds between 58.7% (warm seats) and 84.2% (hot drinks) of each of the nine other kinds, and is right 79.0% to 92.4% of the time when it names one. These names appear on the sensor's screen to explain why something was not counted. They are never counted and never sent.

Speed. On the sensor, owl-4.9 takes 14 to 16 ms per frame on one core of the Pi 5, measured by the sensor's own self test on its own camera. That is fast enough to read every frame the camera sends. An 8-bit quantised export runs 2.1 times faster at a quarter of the size, but it cut owl-3's exact count on the real ceiling set from 94% to 86%. Speed was never the constraint, so the sensor runs full precision.

11 What the experiments taught

Owl went through 16 named versions in four days. Every change was kept or thrown out on the real benchmark sets, and these are the results that shaped it (stress-test figures in this list come from earlier runs and can differ by up to 10 points between draws).

- **Generated frames alone fail on real footage.** owl-2 counted 80% of generated frames exactly, yet on real footage it counted 0% of the ceiling frames and 5% of the side-view frames exactly, having read the texture of a real room as people.
- **Real frames must be mixed in, and a few go a long way.** A 500-step test with real frames mixed in went from 0% to 81% exact on the real ceiling set, and the full run, owl-3, reached 94% on the ceiling set and 74% on the side view.
- **Warmth must be judged against the cool end of the frame.** Taking the room as the frame's 10th percentile instead of its median, which is the person when a person fills the view, lifted the real side view from 74% to 85% exact and from 86% to 98% within one.
- **Measured physics helped, and a bigger network did not.** Measured skin and clothing physics lifted the unseen meeting rooms from 24% to 33% exact and from 58% to 85% within one, while widening the network from 24 to 32 base channels tied the smaller one (94.2% and 73.9% against 93.8% and 74.2% on the ceiling and side view).
- **A sharp target at a guessed head point teaches the model to miss.** On a finer output grid, sharp targets at heads estimated from body boxes dropped the meeting rooms from 97.9% to 67.2% exact, and spreading those targets became standard for every later version.
- **Density counts crowds that peaks merge.** A density head, whose total does not fall when heads touch, lifted a stress-test crowd of ten from 41% to 82% within one with the real benchmark sets unchanged, and became the crowd model.
- **Data shares trade off until the right data arrives.** owl-4.7 bought Flux's camera (88.8% exact against owl-4.2's 75.6%) with the meeting rooms (97.9% down to 82.0%), and adding SDT's rendered Lepton-style frames in owl-4.8 improved both at once, to 90.1% and 93.4% (Figure 9).
- **Validation stopped improving early.** owl-4.9's real validation score stopped rising about a quarter of the way through its 20,000-step run (Figure 10), so the next experiment is new data: more real crowds and side views from this camera.

Two failures belong beside these. owl-4.4 drew more people lying down, more bare torsos and more bodies at the sensor, and lost the real meeting rooms, falling from 97.9% to 60.9% exact with no measurable gain on the cases it targeted, so the change was reverted. Retraining the crowd model starting from owl-4.8 instead of its own earlier weights dropped a crowd of ten to 18.7% exact, so the older crowd model stayed until real crowd data arrived.

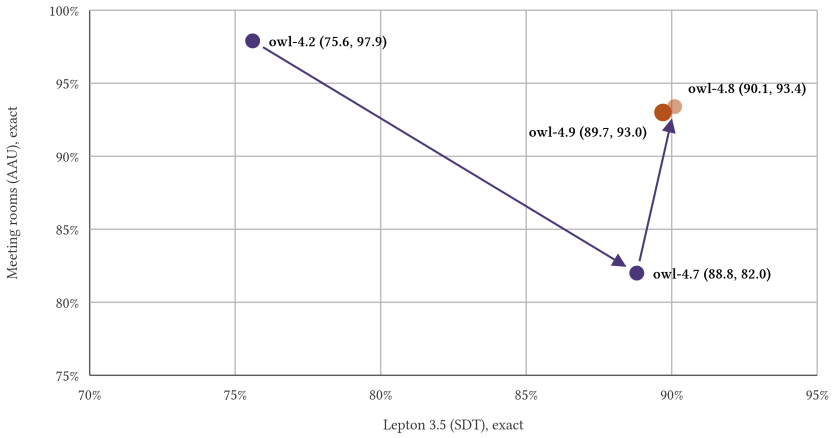


Figure 9. The trade between the Lepton 3.5 frames (SDT, horizontal) and the wall-mounted meeting rooms (AAU, vertical), exact counts. owl-4.2 sits at 75.6% and 97.9%. owl-4.7 bought the camera with the meeting rooms (88.8%, 82.0%). owl-4.8, trained with rendered Lepton frames as well, recovered the meeting rooms without giving the camera back (90.1%, 93.4%). owl-4.9 (89.7%, 93.0%) differs from owl-4.8 by under one point on both sets.

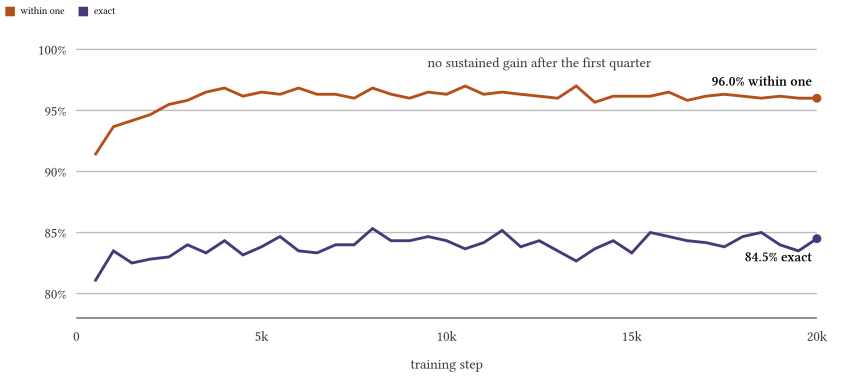


Figure 10. owl-4.9's real validation score every 500 steps over its 20,000-step run, ending at 84.5% exact and 96.0% within one. Both curves are flat after about 5,000 steps. The validation mix includes the new classroom crowds, so these values are not comparable with earlier versions'.

12 Privacy by design

Flux is built to tell people how busy a place is. It must never become a camera that watches them. Owl reads each thermal frame on the device and keeps no history of frames.

Kinds, never who. Owl's outputs are ten kinds of thing. It has no identity output, no face class and no face features. IISER labels faces, and I use each face box only as the point where a head is. Owl is trained to locate people, never to predict who they are.

No memory between frames. Owl reads each thermal frame on its own and keeps no track of anyone. The doorway counter follows a crossing only for the moment it takes to tell an arrival from a departure.

Numbers are all that leave. The network's points and boxes exist only in memory. Boxes are drawn only on units with a screen and are never sent. What leaves the sensor is a handful of numbers: people counts and a sound level, never an image.

Enforced by tests. Tests in the sensor's suite fail the build if Owl gains an identity output or carries anything between frames, or if an image, a point or a box leaves the device.

Data used as licensed. Every dataset is used as its licence or permission allows; the licences and permissions are listed in the acknowledgements. None of their images are redistributed or published, and none are shipped with the model. IISER's licence also forbids using its data to identify anyone, and Owl has no identity output.

The model files are kept private, like Flock's crowd-forecasting model. The generator, network and training and grading code are public. Without a model on disk the sensor counts with the rule, and says so.

13 Conclusion

Owl shows that a model small enough for one core of a Raspberry Pi can count people on a 160 × 120 thermal camera across the cameras, views and warm distractors it was trained on, without recording new footage of anyone to train it and without the sensor keeping a frame after it is counted. Generated frames taught it the warm things the rule confused with people, and licensed real frames taught it what real rooms look like. On five of the six real benchmark sets, owl-4.9 counts 82 to 96% of frames exactly, where the deployed rule manages 9 to 77%, and on a sealed set that guided nothing it counted 94.3% exactly against the rule's 40.7%. On real classroom crowds it misses by one person on average, where its predecessor missed by almost five. The next measurements are real footage from Flux's camera of sparse and crowded rooms, and a first measurement in a working venue.

PART III

How it was built, and what remains

The process from the first collector to the month-long test, the limitations of both studies, and what comes next.

14 How it was built

I built both studies in seven months, from the first collector in March 2026 to the sensor's doorway counter in October. Table 9 gives the timeline. Dates are Eastern time.

14.1 The forecast

The first collector was committed on 10 March 2026 and widened to 34 cities the same day. It gathered weekly rows from the busyness provider, one typical week per venue of 168 hourly values on its 0 to 100 scale, and realtime rows, one reading per venue-hour on a real date. By the last spring reading, on 18 May 2026, the corpus held 3,912,357 rows across 34,784 venues in 34 cities.

The first models predicted busyness directly, with the venue's popular-times value among their inputs. An 80-feature model of this kind scored far higher on validation than on held-out cities: it had learned to copy the curve, and coordinates acted as a lookup table for its five training cities. From then on coordinates were dropped as features, cross-validation held out a whole city at a time, and the label became the *delta*, how far the hour departs from the venue's own curve.

In August I audited the whole pipeline end to end. The model had never loaded in production, because the runtime package that reads an ONNX file was missing, so the rule engine had answered every request. The collector had written no rows since 19 May while reporting success. Both checks now fail loudly. On 12 August v2.3.0 to v2.4 failed their gates and v2.5.0-starling shipped. Over the next six days I made serving build the vector training built, corrected 3,536,360 rows whose stored hour was the provider's array index, fixed the 8 blocking findings of a 21-finding audit, and closed a category-feature leak. The first retrain with every stage checked, v2.6.0-starling, shipped on 18 August: MAE 29.42 against 31.48 for the curve on 67,249 rows in three cities it never saw.

After that, every proposed change faced a gate. v2.7, without month and season, failed. Of about 40 calibrations, a quantile display map was armed on 28 August and left the served number on 26 September, once live readings showed it cost within-one accuracy. Six research passes proposed 42 changes, each attacked by an automated review pass told to refute it; seven survived, and the one shipped was the per-venue trailing offset, which entered model mode at half weight on 6 September and moved MAE from 25.90 to 24.53 on 8,895 September live readings.

Live collection restarted on 1 September for about 1,400 venues in Philadelphia and the Lehigh Valley, with a provenance label on every row, and went hourly on 3 September. From 5 September, Miami, one of the three held-out cities, is read at two fixed hours a day.

On 25 September a replay of the serving code over the 8 September export measured what the card is scored on: whether the served number lands in the same crowd level as the live reading. v2.6.0 with the display map landed within one level 57.4% of the time on 7,920 readings, while a reading one hour old predicted the next within one level 92.0% of the time. Two serving switches followed with no retrain: the curve plus 0.85 of the trailing offset, and a nowcast that weights the latest reading by its

age. Six rounds of adversarial review by automated review tools checked them, and one caught that the offset could see the hour it was scoring. They went live at 18:55 on 26 September. On 8 October a fresh export scored them at 54.9% exact on 119,939 readings over 30 dates, against 56.0% on the three days they were shipped on. On 13 October, scoring the month against persistence baselines showed that a two-hour reading no longer helped, and production switched to a simpler rule: readings up to an hour old at full weight, the corrected curve otherwise. The retrain gate compares against it.

The parity faults of August were the hardest to see, so the safeguards built since are tests that training and serving agree, including a leak test that changes every reading at or after a row's hour and requires none of the row's inputs to move.

14.2 Owl

The sensor began as backend tables and an endpoint on 2 May 2026, tested before any device existed. The demo unit was designed on 4 May from about $1,000 of parts around a Raspberry Pi 5 and a FLIR Lepton 3.5, chosen over the 32 × 24 MLX90640 for its 160 × 120 radiometric image; moving the sensor program onto it on 26 August took the thermal path from 768 pixels to 19,200. On 21 September the device split into a base and a wall head that looks down at a doorway.

Owl replaced the heat-threshold rule in four days, through 16 named versions and 19 full training runs on one RTX 5080. owl-1 (26 September) learned from generated thermal scenes alone; owl-2 and owl-3 (27 September) showed that real frames must be mixed in; owl-4.9 (29 September) added real classroom crowds. Every real frame is used under a commercial licence or with written permission. The shipped pair, owl-4.9 and its crowd model, draws 60% of its training frames from licensed datasets and 40% from the generator.

The doorway counter settled on 25 September on a VL53L8CX time-of-flight sensor that sees the doorway as an 8 × 8 grid of distances and tells arrivals from departures. I wrote its driver on 26 September, before the part arrived, with an occupancy estimate that uses the camera's count as a floor. On 7 October the counter read 265 frames on the board, and the analog microphone was retired in favour of an I2C decibel meter, so the Pi will read a number and never audio. On 8 October the base and the wall head were merged into one box, because the doorway sensor and the decibel meter need short I2C wires.

Table 9. Timeline of both studies, 2026. F: the forecast; C: the counter.

Date	Track	Milestone	Result
10 Mar	F	First collector; 34 cities	Weekly curves and readings begin
Mar to Apr	F	Direct models with the curve as an input	Copied the curve; delta label adopted
2 May	C	Sensor tables and endpoint	Tested before any device existed
4 May	C	Demo unit designed; Lepton 3.5 chosen	About $1,000 of parts
18 May	F	Last spring reading	Corpus of 3,912,357 rows
12 Aug	F	Audit; v2.3 to v2.4 fail; v2.5.0 ships	Model loading and collector checks rebuilt
14 to 16 Aug	F	Parity sweep, clock fix, 21-finding audit, leak closed	3,536,360 rows corrected
18 Aug	F	v2.6.0-starling ships	MAE 29.42 against 31.48, 67,249 rows
19 Aug	F	v2.7 without month and season	Fails gate, MAE 29.89
26 Aug	C	Sensor program moves to the Lepton 3.5	768 to 19,200 pixels
1 Sep	F	Live collection restarts with provenance	One call a second
6 Sep	F	Demand-first sweeps; trailing offset ships	MAE 25.90 to 24.53
25 Sep	F	Band replay; parallel sweeps; retrain rules committed	As served 57.4% within one
26 Sep	F	Curve + offset + nowcast in production, 18:55	Exact level 32.2% to 56.0% on the three-day test
26 Sep	F	Retrain rehearsed with live features	79.7% within one; refused by the gate
27 Sep	C	owl-2 fails on real frames; owl-3 and owl-4.2	Real ceiling 0% to 94%
29 Sep	C	owl-4.9 and its crowd model ship	Crowd error 4.66 to 1.03 people
3 Oct	F	Provider's live coverage drops	Live readings fall to about 61% of the week before
6 Oct	F	Retrain set for 19 to 20 October	
7 Oct	C	Doorway counter's first frames; decibel meter chosen	265 frames
8 Oct	F	Month-long test of the served number	54.9% exact on 119,939 readings, 30 dates
13 Oct	F	Persistence baselines scored; production simplified	Hour-old reading at full weight, else curve + offset
13 Oct	C	Sealed test scored once	94.3% exact on 300 unseen frames; rule 40.7%
14 Oct	F	Retrain moved to early November	Rules unchanged

15 Limitations

- **Relative readings, not headcounts.** Every forecast label is the busyness provider's relative reading on a per-venue 0 to 100 scale, an estimate made by the provider's own system. No hand count or sensor count inside a venue has checked it, though its weekly rhythm tracks the subway entries beside 695 New York venues (median r 0.74, Section 4.10), and a score of 70 means different crowds at different venues. What the forecast predicts is the provider's next live value. Consecutive-hour readings were identical 80.9% of the time in early September and about 45% to

48% from late September, which may reflect how often the provider refreshes its value. On 15 public real-count datasets the provider-specific findings do not hold as stated (Section 4.10), so they describe the provider's estimates rather than crowds in general; 13 of those datasets count people passing a sensor, and the two that count people inside side with the provider's readings on the first finding, as do three more people-inside sets added afterwards (Section 4.10).

- **The test window.** The served accuracy rests on 119,939 readings over 30 dates (9 September to 8 October) in Philadelphia, the Lehigh Valley and Miami: three of the 34 cities, in one season, and the nowcast's gain comes from the two Pennsylvania regions alone. The online correction went into production on a three-day test, before the five-date band gate a model artifact faces; the 30-day test confirms it, and the retrain test is the first pre-registered one. Production's current rule was identified on this window, so the retrain's held-out window is its first out-of-sample test. On *Packed* readings the served number is right far more often when it says *Packed* than the configuration it replaced (67.4% against 28.0%), at slightly lower recall (45.3% against 49.2%, on 20,445 readings).
- **The delta model's training data and its holdout.** The model results of Section 4.7 rest on spring rows of unknown provenance, and its three held-out cities also scored the versions that failed before it, so they act as a validation set. On verified live readings the spring-trained model loses to the curve: over the 30-day test, MAE 30.93 as served (display map on) against 26.40 for the weekly curve, and in the 8 September export the model alone scored 27.52 against 26.42 for the curve on 7,920 readings. Ten of its 106 features are never split on (Section 2.3), latitude re-enters through daylight hours, and the learner comparison is untuned and confined to three cities.
- **Owl in the field.** Owl has not yet been measured in a working venue; the venue study below does that. The six real sets also served as development benchmarks, and the sealed test shows one or two people in an office, from the lab whose side-view set Owl trained on; its counts, like the IISER recounts, are model-made visual counts of the frames rather than counts taken in the room. On cameras it was not trained on, Owl transfers unevenly: the rule beat it on a second 32 × 24 sensor, and outdoors it counted a person in most frames showing only a dog. The one licence-clean public set of indoor groups of three or more, from a 32 × 32 ceiling sensor, went the same way (sealed test 3); Lepton-class sets of indoor groups, including footage from a working restaurant, sit behind registration or need their owners' permission, which has been requested. The hardest cases remain people hidden by people (a side view of three to five people is exact on 40% of frames, 85.5% within one) and sparse classrooms, where Owl counts some warm seat backs. SDT's Lepton frames hold one person or none.
- **No Flux is installed at a venue.** The one unit is a demonstration unit, Flock's crowd number reads no sensor data, and the doorway counter has run on the bench alone.

16 What comes next

Both studies serve one number in Flock: how busy a place will be when a group arrives. The forecast now delivers that number from the venue's own recent readings, and Owl counts people on a device small enough to mount in a doorway. None of the measurements below has been run, and this paper gives no results from them.

Counts as ground truth. The forecast predicts busyness normalised per venue, so a score of 70 could mean a dozen people in a small bar or two hundred in a large one. A Flux unit in a venue that Flock already reads would give that venue an absolute scale. Its headcount, paired with the busyness provider's live score for the same hours, would show how many people a given score means at that venue and whether the five crowd levels match what a group sees at the door. Once proven on a real door, the doorway counter would add arrivals and departures, so the sensor could estimate how many people are inside as well as how many are in view.

Fresher readings. The served number's strength is the venue's latest reading, and its value is concentrated in the first hour. Today's readings are an hour old at best and often several hours old or missing. A sensor that reports every 30 seconds would make that reading minutes old, and on real counts like Flux's the replication favours carrying its departure onto the curve.

The pre-registered retrain. The retrain is set for early November, after Halloween, with pre-registered rules and a held-out window. I will update this paper with its results, whichever way the gate decides.

A harder sealed test for Owl. The sealed test reported here is one or two people against a plain wall. The next one is recorded on Flux's own camera, with consent, in a real room holding from one to fifteen people, counted in the room as it is filmed and kept unseen until the model it grades is final. Sparse rooms, where Owl counts warm seat backs, are the case it must answer first.

The venue study. I would install one unit, with the owner's agreement, at a venue the hourly collector already reads. Hand counts timed against Owl's counts on several busy nights would score Owl in that room before its counts are used as ground truth, and hand counts of arrivals and departures would score the doorway counter. Then, hour by hour over several weeks, I would compare the busyness provider's live score against Flux's count, the score per person, and how often the crowd level names what the count says. The counts are a separate test and change none of the retrain's rules.

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Appendix: Choices made after seeing data

Hypotheses H1 to H3 were written after the preliminary findings on the curve and several of the result analyses had been run; the pre-registered retrain is the first test fixed in advance. The persistence measurements cover 2026-09-01 to 09-08 and so overlap the three test days. They informed the decision to add a nowcast, while its weights were fitted on 09-01 to 09-05. The 30-day window was scored after the three-day result was known. Its boundaries are the day after that test and the date of the export, and every model-served live reading between them is counted. The serving clamp, the one-point push that model mode gives rounded scores outside 25 to 65 toward the nearer end of the scale, the display map, the offset weight and every nowcast weight were chosen on a split earlier than the rows that report them. Weekly rows carried the weather of their collection moment, a leak cleared from 3,246,456 rows. Of rows recording “no event nearby”, 59.5% were schema defaults correlated with city (56.1% of the whole corpus), which is why the coverage comparison disables the events lookup for both arms. The persistence baselines, the two-way bootstrap, the age-mix standardisation, the readings whose value changed and the test-window stickiness figures were computed on 13 October, after the first version of this paper; rule (b) was identified on the 30-day window, has served production since 13 October, and has no out-of-sample figure yet. The three held-out cities scored several model versions

before v2.6.0 passed, as Section 4.7 states. The external replication's plan and its two amendments, which added datasets and horizons, were each hashed before their data were downloaded; its deviations are recorded with it, and its curve-plus-latest-departure predictor was registered without a pass rule.

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The public counts of Section 4.10 come from Brisbane City Council; the City of Melbourne Pedestrian Counting System; Dublin City Council and the National Transport Authority; Cork City Council; Dún Laoghaire-Rathdown County Council (via Smart Dublin); Leeds City Council (Data Mill North, Open Government Licence v3.0); Heart of the City, Auckland; Kanton Basel-Stadt; Stadt Zürich (Open Data Zürich, measurements by hystreet.com); the Cambridge Water Department; the Seattle Department of Transportation; the Metropolitan Transportation Authority (data.ny.gov); Crowdedness at the Campus Gym by Nick Rose (Kaggle, ODbL v1.0); and the KIT Library, Karlsruhe Institute of Technology, whose Seatfinder occupancy data are used with its permission. The room data come from ROBOD (Tekler, Ono, Peng, Zhan, Lasternas and Chong, National University of Singapore; figshare, CC BY 4.0). The campus Wi-Fi data come from Universitat Politècnica de Catalunya, Space and Time User Distribution in a University Campus (Mendeley Data, CC BY 4.0). The ceiling-sensor frames of the third sealed test come from Klir et al., Multi-Person Localization based on a Thermopile Array Sensor with Machine Learning and a Generative Data Model (GitHub, GPL-3.0). Except where noted, they are used under CC BY 4.0 or a public-domain dedication, and none is redistributed with this paper.

Venue busyness readings come from a commercial venue-busyness provider, which has no part in and does not endorse this work. Weather data is provided by OpenWeather (openweathermap.org) under the Open Database License (ODbL). Sports schedules are from TheSportsDB, and event listings are from Ticketmaster. No dataset from these sources is released with this paper.

Every number behind Section 4.10 and Owl's sealed tests is kept in Flock's database, in four research tables apart from the app's own: the 5,735,155 hourly public counts with each source's licence and the attribution it requires, the 1,098 counted frames with Owl's and the rule's count beside each, and the paper's 74 headline numbers, each with the file that computes it. The retrain and the venue study join against these tables, so every comparison in a later version of this paper starts from the same rows.